PHYSICS

1) A 10 kg stone is suspended with a rope of breaking strength 30 kg-wt. The minimum time in which the stone can be raised through a height 10 m starting from rest is (taking g = 10 m/s^2)

(1) 0.5 second

(2) 1.0 second

(3) $\sqrt{\frac{2}{3}}s$

(4) 2.0 second

2) A ball of mass 400 gm is dropped from a height of 5 m. A boy on the ground hits the ball vertically upwards with a bat with an a average force of 100 N, so that it attains a vertical height of 20 m.The time for which the ball remain in contact with the bat is (g = 10 m/s^2)

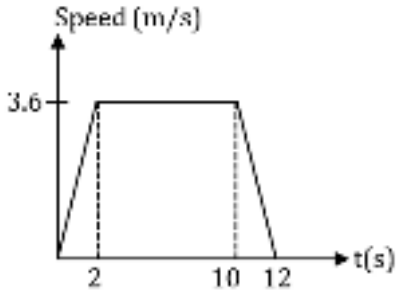
(1) 0.12 second

(2) 0.08 second

(3) 0.04 second

(4) 12 second

3) A lift is going up. The total mass of the lift and the passenger is 1500 kg. The variation in the speed of the lift is as shown in the graph. Find the tension in the rope pulling the lift at t = 11 th second.



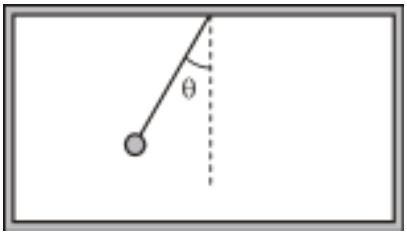
(1) 17400 N

(2) 14700 N

(3) 12000 N

(4) Zero

4) A small ball of mass 800 g is suspended with a light string from the ceiling of cabin. When the cabin moves towards the right with uniform acceleration a, the string settles to an angular position θ as shown. If the tension in the string is 10 N, which of the following statements is/are true? [g = 10

m/s^2] 

(A) θ = 37° (B) θ = 53°
(C) a = 7.5 m/s^2 (D) a = 10 m/s^2

(1) A and B
(2) B and C
(3) A and C
(4) A and D

5) Five forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \vec{F}_4$ and $\vec{F}_5$ are acting on a particle of mass 2.0 kg so that it is moving with 4 m/s^2 in east direction. If force $\vec{F}_1$ is removed, then the acceleration becomes 7 m/s^2 in north, then the acceleration of the block if only $\vec{F}_1$ is acting will be :-

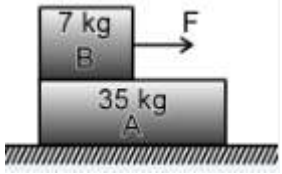
(1) 16 m/s^2
(2) $\sqrt{65}$ m/s^2
(3) $\sqrt{260}$ m/s^2
(4) $\sqrt{33}$ m/s^2

6) A machine gun fires a bullet of mass 40 g with a velocity 1200 m/s. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can be fire per second at the most?

(1) one
(2) Four
(3) Two
(4) three

7) A chain consisting of 5 links, each of mass 0.1 kg if lifted vertically with a constant acceleration of 2.5 m/s^2. The force of interaction between the top link and link immediately below it will be

(1) 6.15 N
(2) 4.92 N
(3) 9.84 N
(4) 2.46 N

8) Block A of mass 35 kg is resting on a frictionless floor. Another block B of mass 7 kg is resting on it as shown in the figure. The coefficient of static friction between the blocks is 0.5 while kinetic friction is 0.4. If a force of 100 N is applied to block B, the acceleration of the block A will be (g = 10 m/s) 

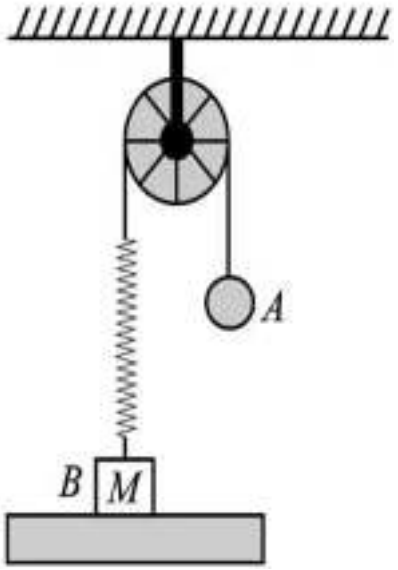
(1) 0.8 m/s^2

(2) 2.4 m/s^2

(3) 0.4 m/s^2

(4) 4.4 m/s^2

9) In the figure, the ball A is released from rest, when the spring is at its natural (unstretched) length. For the block B of mass M to leave contact with ground at some stage, the minimum mass of A must be



(1) 2M

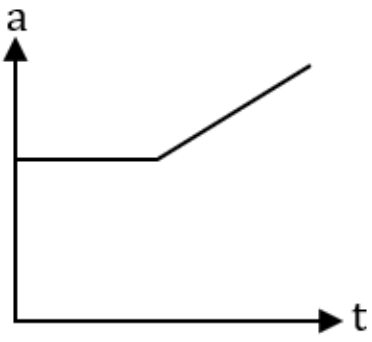
(2) M

(3) M/2

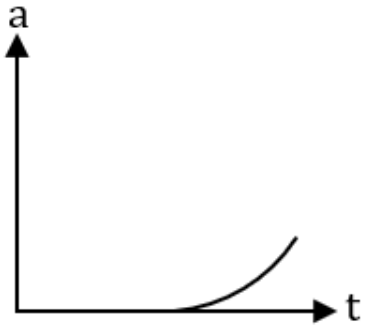
(4) M/4

10) A block is placed on a rough horizontal plane. A time dependent horizontal force F = Kt acts on the block. Here, K is a positive constant. Acceleration-time graph of the block is

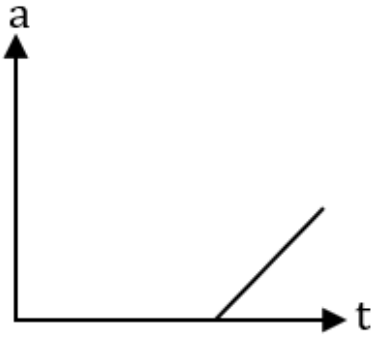
(1)



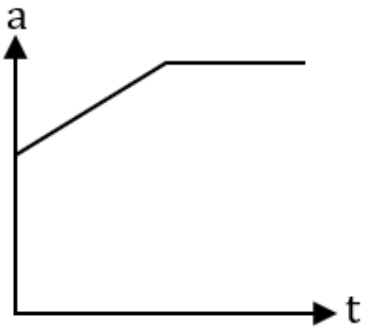
(2)



(3)



(4)



11) A body of mass 3.0 kg moves under the influence of some external force such that its position S as a function of time t is given by $s = 6t^3 - t^2 + 1$ where S is in metres and t is in seconds. The work done (in joule) by the force in first three seconds is :-

(1) 18 J

(2) 1800 J

(3) 3660 J

(4) 36504 J

12) A force $\vec{F} = -k(y\hat{i} + x\hat{j})$, where k is a constant, acts on a particle moving in the x-y plane. Starting from the origin, the particle is taken from the origin (0, 0) to the point (a, 0) and then from the point (a, 0) to the point (a, a). The total work done by the force $\vec{F}$ on the particle is

(1) ka^2

(2) $-ka^2$

(3) $2ka^2$

(4) $-2ka^2$

13)

A body of mass 6 kg is acted upon by a force which causes a displacement in it given by $x = \frac{t^2}{4} m$ where t is the time in second. The work done by the force in 2 second is

(1) 12 J

(2) 9 J

(3) 6 J

(4) 3 J

14) A particle moves along a curve of unknown shape but magnitude of force $\vec{F}$ is constant and always acts along the tangent to the curve. The,

(1) $\vec{F}$ must be conservative

(2) $\vec{F}$ may be conservative

(3) $\vec{F}$ must be non-conservative

(4) $\vec{F}$ may be non-conservative

15) Two springs P and Q are stretched by applying forces of equal magnitudes at the four ends. If the spring constant of P is 2 times greater than that of Q and the energy stored in P is E, then the energy stored in Q is

(1) $\frac{E}{4}$

(2) $\frac{E}{2}$

(3) E

(4) 2E

16) The potential energy of a particle of mass 5 kg moving in the x-y plane is given by U = –7x + 24y joule, x and y being in metre, initially at t = 0 the particle is at the origin and moving with velocity $(2\hat{i} + 3\hat{j})$ m/s. The magnitude of the force on the particle and the acceleration of the particle are

(1) 35 N, 7 m/s^2

(2) 30 N, 6 m/s^2

(3) 25 N, 5 m/s^2

(4) None of these

17) The potential energy of a body is given by $U = 40 + 6x^2 – 7xy + 8y^2 + 32 z$
Where U is in joule and x, y, z in metre. Deduce the x, y and z components of the force (in newton) on the body when it is in position (–2, 0, + 5)

(1) $F_x = – 32$; $F_y = –14$; $F_z = 24$

(2) $F_x = – 14$; $F_y = –32$; $F_z = 24$

(3) $F_x = 24$; $F_y = –32$; $F_z = –14$

(4) $F_x = 24$; $F_y = –14$; $F_z = –32$

18) 10 litres of water per second is lifted from a well through 20 m and delivered with a velocity of

10 m/s, then the power (In kW) of the motor is take g = 10 m/s
(Take g = 10 m/s^2)

(1) 0.5

(2) 1.25

(3) 2.5

(4) 5.0

19) The potential energy of a 1 kg particle free to move along the x-axis is given by $V = \left(\frac{x^4}{4} - \frac{x^2}{2}\right) J$
The total mechanical energy of the particle is 2J. Then the maximum speed is

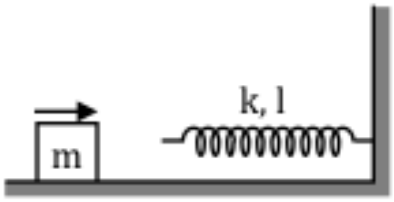
(1) $\frac{3}{\sqrt{2}} m/s$

(2) $\sqrt{2} m/s$

(3) $\frac{1}{\sqrt{2}} m/s$

(4) 2 m/s

20) The block of mass M moving on the frictionless horizontal surface collides with a spring of spring constant k and compresses it by length L. The maximum momentum of the block after collision is



(1) zero

(2) $\frac{ML^2}{k}$

(3) $\sqrt{Mk}\, L$

(4) $\frac{kL^2}{2M}$

21) A body of mass 4 kg is moving with momentum of 8 kg-m/s. A force of 0.2 N acts on it in the direction of motion of the body for 10 s. The increases in KE in joule

(1) 10

(2) 8.5

(3) 4.5

(4) 4

22) A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time t as $a_c = k^2rt^2$, where k is a constant. The power delivered to the particle by the forces acting on it is

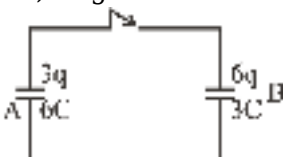
(1) zero

(2) $mk^2r^2t^2$

(3) mk^2r^2t

(4) mk^2rt

23) In given circuit when switch has been closed, then final charge on capacitor A & B respectively



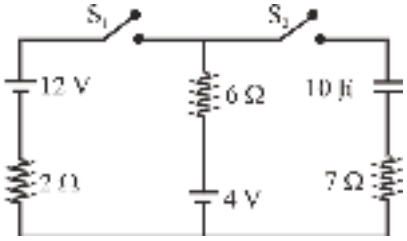
(1) 3q, 6q

(2) 6q, 3q

(3) 4.5q, 4.5q

(4) 5q, 4q

24) In the circuit shown in the figure, if switches S_1 and S_2 have been closed for a long time, then current in 6Ω resistance will be :–



(1) Zero

(2) 1 A

(3) 3 A

(4) 2.1 A

25) A parallel plate capacitor is charged using a battery. After the capacitor is fully charged, the battery is disconnected and the plates are moved apart, then :-

(1) the charge on the plates increases

(2) the potential decreases

(3) the capacitance increases

(4) the electric fields strength remains constant.

26) A capacitor is connected to battery if distance between two plates increases then :

(1) Charge on plates remains same.

(2) Potential difference increases.

(3) Stored energy decreases.

(4) Electric field remains same.

27) A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V. Another capacitor of capacitance 2C is connected to another battery and is charged to potential difference 2V. The charging batteries are now disconnected and the capacitors are

connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is

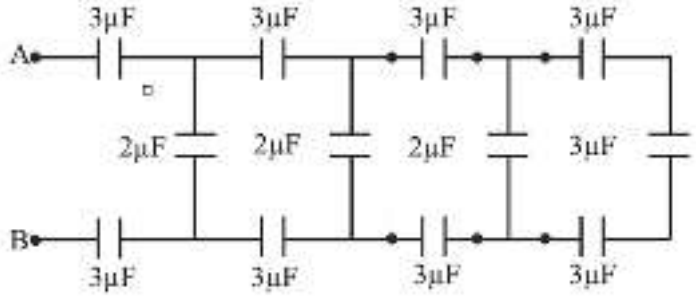
(1) zero

(2) $\frac{25CV^2}{6}$

(3) $\frac{3CV^2}{2}$

(4) $\frac{9CV^2}{2}$

28) The resultant capacitance between A and B in the figure is :



(1) 1μF

(2) 10 μF

(3) 50 μF

(4) 1.5

29)

If the potential of a capacitor having capacity 6 μF is increased from 10 V to 20 V, then increase in its energy will be :-

(1) 12×10^{-4}J

(2) 4×10^{-4}J

(3) 9×10^{-4}J

(4) 12×10^{-6}J

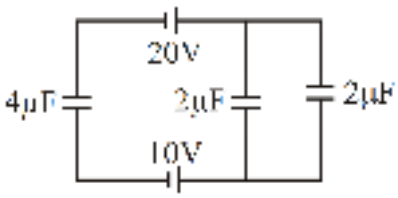
30) A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2ms. The power delivered to the patient is :-

(1) 45 kW

(2) 90 kW

(3) 180 kW

(4) 360 kW



31) In the circuit shown in figure, charge stored in 4μF capacitor is :-

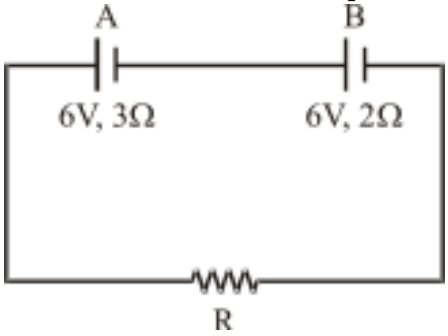
(1) 20 μC

(2) 10 μC

(3) 40 μC

(4) 120 μC

32) Two sources of emf 6 V and internal resistance 3 Ω and 2 Ω are connected to an external resistance R as shown. If potential difference across battery A is zero, then value of R is :



(1) 1 Ω

(2) 2 Ω

(3) 3 Ω

(4) 4 Ω

33) A square plate of side 'a' and circular plate of radius 'a/2' are placed coaxially and parallel to each other to form a parallel plate capacitor of capacitance 'C' then separation between the plates is :-

(1) $\frac{a^2\epsilon_0}{C}$

(2) $\frac{\pi a^2\epsilon_0}{C}$

(3) $\frac{\pi a^2\epsilon_0}{4C}$

(4) $\frac{\epsilon_0}{C}a^2\left(1-\frac{\pi}{4}\right)$

34) A slab of copper of thickness $\frac{d}{3}$ is introduced between the plates of a parallel plate capacitor where d is the separation between its two plates. If the capacitance of the capacitor without the copper slab is C and with copper slab is C' then $\frac{C'}{C}$ is :-

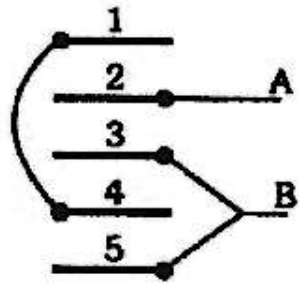
(1) $\sqrt{2}$

(2) 2

(3) $\frac{3}{2}$

(4) $\frac{1}{\sqrt{2}}$

35) Five identical metal plates 1, 2, 3, 4 and 5 each of area A on one side are fixed parallel and equidistant (d) to each other. The plates 1 and 4 are joined by a conductor and plates 3 and 5 are also joined by a conductor as shown in figure. Then, the capacitance of this system between A and B



is :

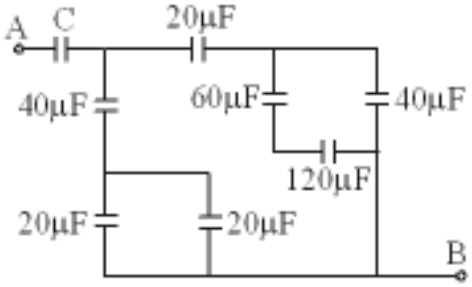
(1) $\frac{5\varepsilon_0 A}{d}$

(2) $\frac{4\varepsilon_0 A}{d}$

(3) $\frac{5\varepsilon_0 A}{3d}$

(4) None

36) Figure shows a network of capacitors. Find the value of C, if the effective capacitance between



the points A and B is 12 μF.

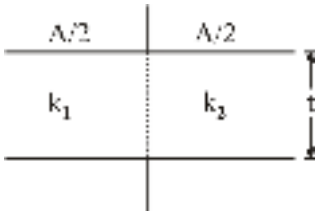
(1) 18 μF

(2) 20 μF

(3) 10 μF

(4) 5 μF

37) A parallel plate condenser is filled with two dielectrics as shown. Area of plate is A metre^2 and the separation is t metre. The dielectric constants are k_1 and k_2 respectively. Its capacitance in farad



will be :-

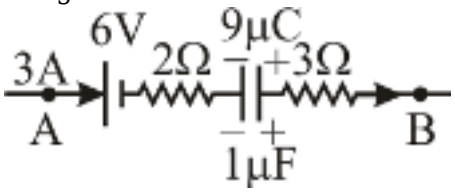
(1) $\frac{\varepsilon_0 A}{t}(k_1 + k_2)$

(2) $\frac{\varepsilon_0 A}{t} \cdot \frac{k_1 + k_2}{2}$

(3) $\frac{2\varepsilon_0 A}{t}(k_1 + k_2)$

(4) $\frac{\varepsilon_0 A}{t} \cdot \frac{k_1 - k_2}{2}$

38) The potential difference V_A–V_B between points A and B for the circuit segment shown in figure at the given instant is



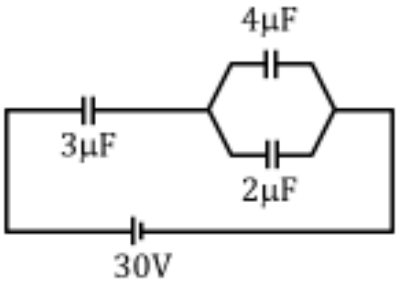
(1) 12 V

(2) –12 V

(3) 6 V

(4) –6 V

39) In the circuit shown in figure, match the following two columns.



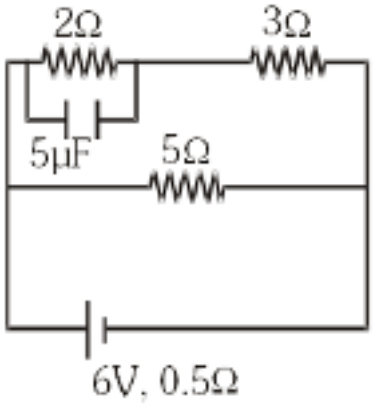
Column-I	**Column-II**
(I) Charge given by battery	(a) 20 μC
(II) Charge of 4μF capacitor	(b) 30 μC
(III) Charge of 2 μF capacitor	(c) 60 μC
(IV) Charge of 3 μF capacitor	(d) 40 μC

(1) I → a ; II → b ; III → b ; IV → c

(2) I → c ; II → d ; III → a ; IV → c

(3) I → b ; II → a ; III → c ; IV → d

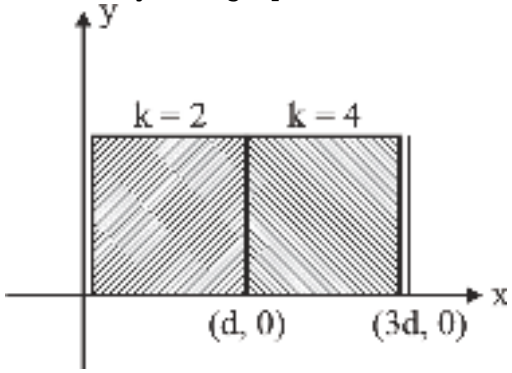
(4) I → c ; II → d ; III → c ; IV → a

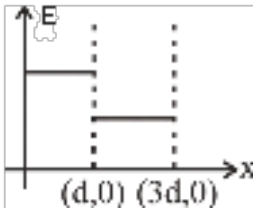


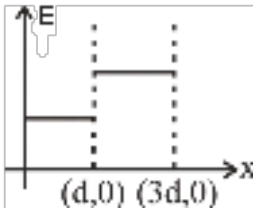
40) Find out amount of charge on capacitor shown in circuit.

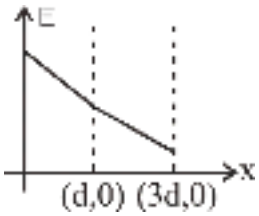
(1) 0μC
(2) 30μC
(3) 10μC
(4) 25μC

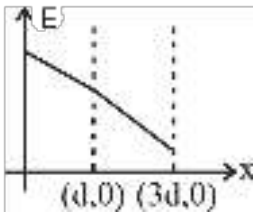
41) A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) and distance (x) from left plate :



(1)


(2)


(3)


(4) 

42) The capacity of an isolated sphere is increased n times when it is enclosed by an earthed concentric sphere. The ratio of their radii is

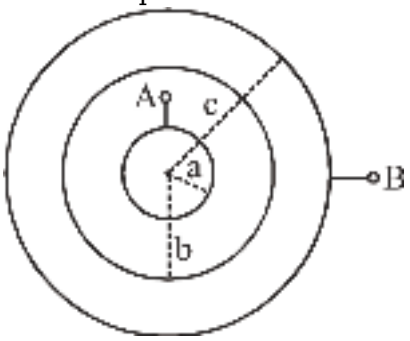
(1) $\frac{n^2}{n-1}$

(2) $\frac{n}{n-1}$

(3) $\frac{2n}{n+1}$

(4) $\frac{2n+1}{n+1}$

43) Consider an assembly of three conducting concentric spherical shells of radii a, b and c as shown. Find the capacitance between A and B :-



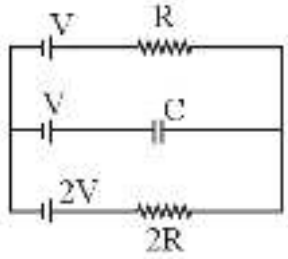
(1) $\frac{4\pi\varepsilon_0 ac}{c-a}$

(2) $\frac{4\pi\varepsilon_0 ab}{b-a}$

(3) $\frac{4\pi\varepsilon_0 bc}{c-b}$

(4) $\frac{4\pi\varepsilon_0 abc}{(c-b-a)}$

44) In the circuit of figure with steady current, the potential drop across the capacitor is :-

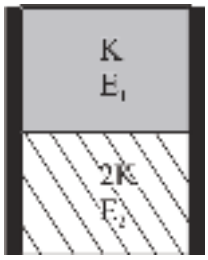


(1) V

(2) $\frac{V}{2}$

(3) $\frac{V}{3}$

(4) $\frac{2V}{3}$

45) A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and 2K are now introduce in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then 

(1) $E_1 = 2E_2$

(2) Electrostatic potential energy of upper half is less than that of lower half

(3) Induced charges on both slabs are same

(4) Charge distribution on the plates remains same after insertion of dielectric

CHEMISTRY

1) The correct order of ionic radii for the following ions is:

(1) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$

(2) $Cr^{3+} < Mn^{3+} < Ti^{3+} < V^{3+}$

(3) $Mn^{3+} < Cr^{3+} < V^{3+} < Ti^{3+}$

(4) $Ti^{3+} > V^{3+} > Cr^{3+} > Mn^{3+}$

2) Lanthanoid contraction is a consequence of:

(1) Shielding of 4f electrons by 5d electrons.

(2) Poor shielding of 4f electrons by each other.

(3) Increase in the number of 5d electrons.

(4) Decrease in the number of 5d electrons.

3) The general electronic configuration of d-block elements is:

(1) $(n-1)d^{1-10}ns^{1-2}$

(2) $(n-1)f^{1-14}ns^{0-2}$

(3) $(n-1)d^{1-10}ns^{0-4}$

(4) $nf^{1-14}ns^{1-2}$

4) Which of the following ions is colorless?

(1) Sc^{3+}

(2) Ti^{3+}

(3) V^{3+}

(4) Cr^{3+}

5) Which of the following statements about d-block elements is incorrect?

(1) They form colored ions.

(2) They exhibit variable oxidation states.

(3) They are generally poor catalysts.

(4) They are mostly paramagnetic in nature.

6) Which of the following is the most stable ion in an aqueous solution?

(1) Cr^{2+} (d^4)

(2) V^{2+} (d^3)

(3) Fe^{3+} (d^5)

(4) Co^{2+} (d^7)

7) Which of the following has large positive value of $E^{\circ}_{M^{3+}/M^{2+}}$?

(1) Ti

(2) V

(3) Cr

(4) Mn

8) The transition element which does not exhibit variable oxidation states is:

(1) Sc

(2) Ti

(3) Co

(4) Ni

9) Which of the following aquated metal ion is not colourless?

(1) Sc^{3+}

(2) Zn^{2+}

(3) Ti^{4+}

(4) V^{3+}

10) The maximum oxidation state shown by Mn in its compound is (with only F)

(1) +4

(2) +5

(3) +6

(4) +7

11) Highest oxidation state of manganese in fluoride is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7) because _____.

(1) fluorine is more electronegative than oxygen.
(2) fluorine does not possess *d*-orbitals.
(3) fluorine stabilises lower oxidation state.
(4) in covalent compounds fluorine can form single bond only while oxygen forms double bond.

12) Which of the following is the electronic configuration of Cm (Z = 96)

(1) [Xe] $4f^7\ 5d^1\ 6s^2$
(2) [Rn] $4f^7\ 5d^1\ 6s^2$
(3) [Rn] $5f^7\ 5d^1\ 7s^2$
(4) [Rn] $5f^7\ 7s^2$

13) Which of the following compounds has a three-dimensional polymeric structure?

(1) B_2H_6
(2) SiO_2
(3) $AlCl_3$
(4) S_8

14) The correct order of Lewis acidic strength for the boron halides is:

(1) $BF_3 > BCl_3 > BBr_3 > BI_3$
(2) $BI_3 > BBr_3 > BCl_3 > BF_3$
(3) $BCl_3 > BF_3 > BBr_3 > BI_3$
(4) $BBr_3 > BI_3 > BCl_3 > BF_3$

15) Which of the following allotropes of carbon is used as a lubricant?

(1) Diamond
(2) Graphite
(3) Fullerenes
(4) Carbon black

16) The correct order of thermal stability of the hydrides of Group 16 elements is:

(1) $H_2O < H_2S < H_2Se < H_2Te$
(2) $H_2O > H_2S > H_2Se > H_2Te$
(3) $H_2Te > H_2Se > H_2S > H_2O$
(4) $H_2S > H_2O > H_2Se > H_2Te$

17) In which of the following compounds does fluorine exhibit an oxidation state other than -1?

(1) F_2
(2) HF
(3) NaF
(4) CaF_2

18) The most common oxidation state of the elements of Group 13 is:

(1) +1
(2) +2
(3) +3
(4) +4

19) Which of the following hydrides of Group 15 elements is the strongest reducing agent?

(1) NH_3
(2) PH_3
(3) BiH_3
(4) AsH_3

20) The oxidation state of phosphorus in hypophosphorous acid (H_3PO_2) is:

(1) +1
(2) +3
(3) +5
(4) +2

21) Which of the following represents the correct order of bond dissociation enthalpy for halogens?

(1) $F_2 > Cl_2 > Br_2 > I_2$
(2) $Cl_2 > Br_2 > F_2 > I_2$
(3) $I_2 > Br_2 > Cl_2 > F_2$
(4) $F_2 > I_2 > Br_2 > Cl_2$

22) Which of the following is not an oxoacid of phosphorus?

(1) Hypophosphorous acid
(2) Pyrophosphoric acid
(3) Metaphosphoric acid
(4) Peroxodisulphuric acid

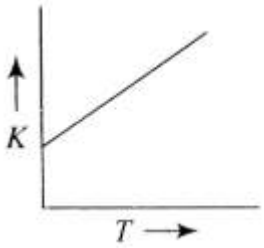
23) Which statement about molecularity of a reaction is wrong ?

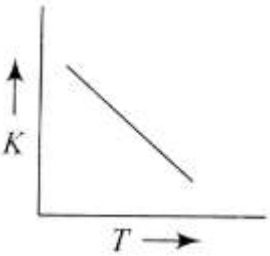
(1) It is the number of molecules of the reactants taking part in a single step of reaction
(2) It is calculated from the reaction mechanism

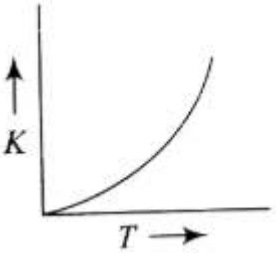
(3) It may be either whole number or fractional

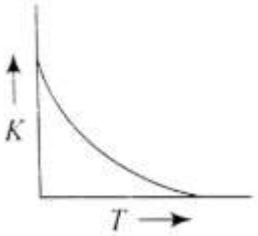
(4) None of the above

24) Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is:

(1) 

(2) 

(3) 

(4) 

25) Which of the following statement is correct for a reaction X + 2Y $\longrightarrow$ Products?

(1) The rate of disappearance of X = twice the rate of disappearance of Y

(2) The rate of disappearance of X = 1/2 rate of appearance of products

(3) The rate of appearance of products = 1/2 the rate of disappearance of Y

(4) The rate of appearance of products = 1/2 the rate of disappearance of X

26) The chemical reaction, $2O_3 \rightarrow 3O_2$ proceeds as follows:

$O_3 \rightleftharpoons O_2 + O$ ...(Fast)

$O + O_3 \longrightarrow 2O_2$ ...(Slow)

The rate law expression should be :

(1) $r = k[O_3]^2$

(2) $r = k[O_3]^2[O_2]^{-1}$

(3) $r = k[O_3][O_2]$

(4) unpredictable

27) The rate law of the reaction, $A + 2B \rightarrow Product.$ Product is given by $\frac{d[P]}{dt} = k[A]^2.[B].$ If A is taken in large excess, the order of the reaction will be :

(1) Zero

(2) 1

(3) 2

(4) 3

28) For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, the are $\frac{d[NH_3]}{dt} = 2 \times 10^{-4} Ms^{-1}$. Therefore, the $-\frac{d[N_2]}{dt}$ rate is given as:

(1) 10^{-4} M s^{-1}

(2) 10^{4} M s^{-1}

(3) 10^{-2} M s^{-1}

(4) 10^{-4} s M^{-1}

29) The rate constant k_1 and k_2 for two different reactions are 10^{16} $e^{-2000/T}$ and 10^{15} $e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is :

(1) $\frac{2000}{2.303}K$

(2) $2000\,K$

(3) $\frac{1000}{2.303}K$

(4) $1000\,K$

30) Half-life period of a first order reaction is 1386 seconds. The specific reaction rate of the reaction is :

(1) 5.0×10^{-2} s^{-1}

(2) 5.0×10^{-3} s^{-1}

(3) 5×10^{-5} s^{-1}

(4) 0.5×10^{-3} s^{-1}

31) During the kinetic study of the reaction $2A + B \rightarrow C + D$ following results were obtained.

Run	[A] in M	[B] in M	Initial rate in M s^{-1}
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

On the basis of above data which one is correct ?

(1) $r = k[A]^2[B]$

(2) $r = k[A][B]$

(3) $r = k[A]^2[B]^2$

(4) $r = k[A][B]^2$

32) In question given below a statement **(S)** and explanation **(E)** is given. Choose the correct answers from the codes (a), (b), (c) and (d) given for each question:
S: Every collision between molecules leads to a chemical reaction.
E: Only those molecules react during collisions which acquire threshold energy level and have proper orientation

(1) **S** is correct but **E** is wrong.

(2) **S** is wrong but **E** is correct

(3) Both **S** and **E** are correct and **E** is correct explanation of **S.**

(4) Both **S** and **E** are correct but **E** is not correct explanation of **S**.

33) In a first order reaction A → B B, if k is the rate constant and initial concentration of the reactant is 0.5 M, then half-life is :

(1) $\frac{\ln 2}{k}$

(2) $\frac{\ln 2}{k\sqrt{0.5}}$

(3) $\frac{\log_{10} 2}{k}$

(4) $\frac{0.693}{0.5\,k}$

34) Activation energy of a chemical reaction can be determined by :

(1) evaluating rate constants at two different temperatures

(2) evaluating velocities of reaction at two different temperatures

(3) evaluating rate constant at standard temperature

(4) changing concentration of reactants

35) The concentration of pollutant in ppm (w/w), that has been measured at 450 mg per 150 kg of sample is:

(1) 3 ppm

(2) 6 ppm

(3) 3000 ppm

(4) 330 ppm

36) Benzene (C_6H_6) and toluene ($C_6H_5CH_3$) form nearly ideal solutions. At 20°C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20°C for a solution containing 78 g of benzene and 46 g of toluene in torr is :

(1) 50

(2) 25

(3) 37.5

(4) 53.5

37) 1 mole of heptane (v. p = 92 mm of Hg) was mixed with 4 moles of octane (v. p = 31 mm of Hg) The vapour pressure of resulting ideal solution will be:-

(1) 43.2

(2) 93

(3) 28

(4) 30

38) A solution is defined as a :

(1) homogeneous mixture of two or more substances

(2) heterogeneous mixture of two or more substances

(3) homogeneous mixture of liquid and solid components only

(4) homogeneous mixture consisting of water as one of the components

39) Correct statements among the following are
(a) Commercial bleaching solution contains 3.62 mass percentage of sodium hypochlorite in water

(b) 35 % (v/v) solution of ethylene glycol, an antifreeze, is used in cars for cooling the engine
(c) Sum of mole fraction of all components in a solution is one
(d) Mass % and mole fraction are independent of temperature

(1) (a) and (c)

(2) (a) and (b)

(3) (a), (b) and (d)

(4) (a), (b), (c) and (d)

40) 0.59 gm. of a dibasic acid is present in 100 c.c. of $\frac{N}{10}$ solution. What is the molecular weight of the acid ?

(1) 59

(2) 118

(3) 100

(4) 11.8

41) In osmosis phenomenon net flow of :

(1) Solvent molecules occurs from higher concentration to lower concentration

(2) Solvent molecules occurs from lower concentration to higher concentration

(3) Solute molecules occurs from higher concentratioin to lower concentration

(4) Solute molecules occurs from lower concentration to higher concentration

42) Vapour pressure of pure A-p_A^o = 100 mmHg
Vapour pressure of pure B-p_B^o = 150 mmHg
A solution containing 2 moles A and 3 moles B will have total vapour pressure, approximately :-

(1) 130 mm Hg
(2) 135 mm Hg
(3) 140 mm Hg
(4) 145 mm Hg

43) For a binary ideal liquid solution, the total pressure of the solution is given as :-

(1) $P_{total} = P_A^o + (P_B^o - P_A^o) X_B$
(2) $P_{total} = P_A^o + P_B^o$
(3) $P_{total} = P_B^o + (P_B^o - P_A^o) X_A$
(4) $P_{total} = P_B^o + (P_B^o - P_A^o) X_B$

44) Which of the following solutions at the same temperature will be isotonic :

(1) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in one litre water.
(2) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in 0.1 litre water.
(3) 3.42 g of cane sugar in one litre water and 0.585g of NaCl in one litre water.
(4) 3.42 g of cane sugar in one litre water and 1.17 g of NaCl in one litre water.

45) What would be the freezing point of aqueous solution containing 18 g of $C_6H_{12}O_6$ in 1000 g of water ? (K_f = 1.86 K molality^{-1})

(1) –0.186° C
(2) –0.372° C
(3) –0.54° C
(4) –0.72° C

BIOLOGY

1) Living organisms exhibit distinctive characteristics to differentiate from non living objects. Which one of the following statements can be considered as defining property of living things ?

(1) Decrease in body mass as a criterion for growth.
(2) Reproduction is synonymous without growth in unicellular organisms.
(3) Ability of living organisms to sense their surroundings.
(4) Growth is found in living organisms only.

2) Which of the following is a defining characteristic of living organisms?

(1) Growth

(2) Ability to make sound

(3) Reproduction

(4) Response to external stimuli

3)

In the given columns, column I contains the terms and column II contains its description. Select the correct match from the option given below.

Column I		Column II	
A.	Growth	I.	Production of offspring
B.	Reproduction	II.	Composed of one or more living cells
C.	Metabolism	III.	Increase in mass and increase in number of individuals
D.	Cellular organization	IV.	Sum total of all chemical reactions occurring in a living body

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-IV, C-III, D-I

4) Which of the following is a unique feature of living organisms?

(1) Growth and reproduction.

(2) Reproduction and ability to sense environment.

(3) Metabolism and consciousness.

(4) All of the above

5)

Which two points are known as the twin characteristics of growth?
(i) Increase in mass
(ii) Differentiation
(iii) Increase in number of individuals
(iv) Response to stimuli

(1) (i) and (ii)

(2) (i) and (iv)

(3) (ii) and (iii)

(4) (i) and (iii)

6) Who is known as "The Darwin of 20th century -

(1) Carl Linnaeus

(2) Joseph Priestley

(3) Ernst Mayr

(4) Julius von Sachs

7) Growth occurs _____ in plants and _____ in animals.

(1) Continuously, only up to a certain age
(2) Only up to a certain age, continuously
(3) Continuously, never
(4) Once, twice

8) Defining characters of all livings is/are :-

(1) Metabolism
(2) Consciousness
(3) Cellular organization
(4) All of the above

9)

Read the following six statements regarding taxonomic categories and choose the correct option.
(A) An order is an assemblage of related families that show a few similar characters.
(B) The number of similar characters increases as we move from species to kingdom.
(C) Convolvulaceae and Solanaceae are placed under the order Polymoniales mainly based on floral characters.
(D) The order Carnivora includes families such as Felidae and Canidae.
(E) Classes including fishes, amphibians, reptiles, birds, and mammals are placed under the phylum Chordata due to the presence of a notochord and a dorsal hollow neural system.
(F) Kingdom is the lowest taxonomic category in biological classification.

(1) A, C, D and E are correct
(2) A, B, C and F are correct
(3) B, D, E and F are correct
(4) A, C and F are correct

10) Match **Column I** with **Column II** and choose the correct option:

	Column I		**Column II**
a.	Class	i.	Plantae
b.	Order	ii.	Mammalia
c.	Family	iii.	Carnivora
d.	Kingdom	iv.	Felidae

(1) (a)–ii, (b)–iii, (c)–iv, (d)–i
(2) (a)–iii, (b)–iv, (c)–ii, (d)–i
(3) (a)–iv, (b)–ii, (c)–iii, (d)–i
(4) (a)–ii, (b)–i, (c)–iii, (d)–iv

11) Cyanobacteria are ecologically significant primarily because they:

(1) Fix atmospheric nitrogen exclusively.

(2) Were among the earliest oxygen-evolving organisms.

(3) Form symbiotic associations with fungi.

(4) Are primary decomposers.

12) Which of the following statements correctly apply to fungal digestion?
A. Enzymes are secreted outside the body
B. Digestion occurs extracellularly
C. Digested nutrients are absorbed
D. Food is engulfed by exocytosis

(1) A, B and C only

(2) A and D only

(3) B and D only

(4) All four

13) Which of the following statements about protists is correct?

(1) All protists are multicellular.

(2) Slime moulds produce sporangia during favorable conditions.

(3) Dinoflagellates have a cellulose-based cell wall.

(4) Euglenoids have peptidoglycan in their cell walls.

14) Members of Phycomycetes are found in:
(i) Aquatic habitats
(ii) On decaying wood
(iii) Moist and damp places
(iv) As obligate parasites on plants

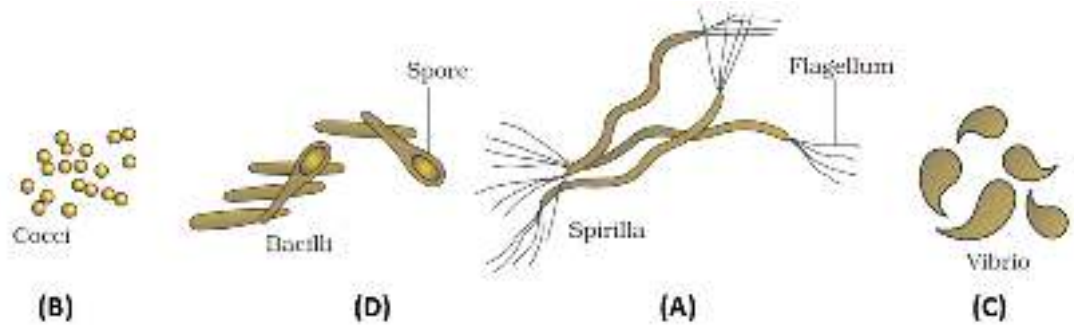
(1) i and ii only

(2) i, ii and iii only

(3) All of the above

(4) None of the above

15)



Identify the correct match between bacterial shape and example.

Example	Shape

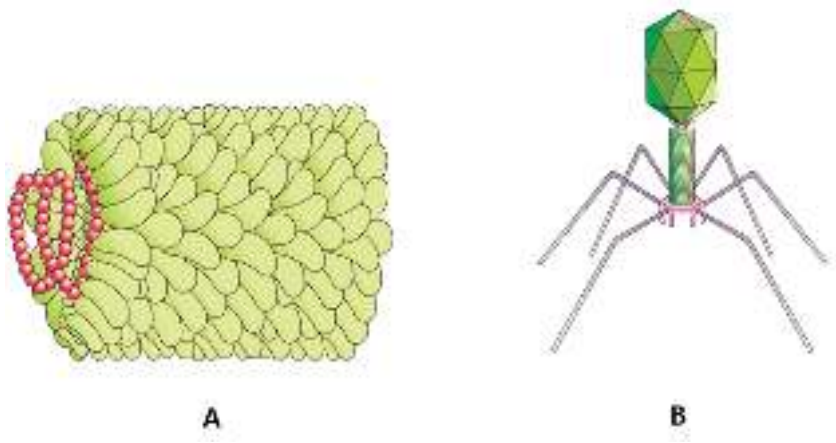
A: Spirillum	1. Helically twisted bacteria
B: Vibrium	2. Comma-shaped
C: Bacillus	3. Rod-shaped
D: Coccus	4. Spherical

(1) A–1, B–2, C–3, D–4

(2) A–3, B–4, C–1, D–2

(3) A–2, B–1, C–4, D–3

(4) A–4, B–3, C–2, D–1



16)
Which feature is absent in both viruses shown?

(1) Capsid

(2) Cell membrane

(3) Nucleic acid

(4) Ability to infect living cells

17) Which of the following organisms can survive extreme environmental conditions due to unique structural adaptations?

(1) Halophilic bacteria

(2) N_2 fixing bacteria

(3) *Mycoplasma*

(4) Dinoflagellates

18) **Assertion (A):** Fungi are grouped separately from plants.
Reason (R): Fungi are heterotrophs with chitinous cell walls, unlike the autotrophic green plants.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, R is false.

(4) A is false, R is true.

19) Match the fungal class with key features:

A	Phycomycetes	1	Sac-like asci

B	Ascomycetes	2	Basidiospores on basidia
C	Basidiomycetes	3	Aseptate coenocytic hyphae
D	Deuteromycetes	4	Imperfect fungi (no sexual stage)

(1) A–3, B–1, C–2, D–4

(2) A–1, B–2, C–3, D–4

(3) A–4, B–3, C–1, D–2

(4) A–2, B–4, C–1, D–3

20) Which of the following statements is incorrect?

(1) Euglenoids lack a cell wall and have a pellicle.

(2) Viroids contain RNA but no protein coat.

(3) *Penicillium* belongs to Deuteromycetes.

(4) Mycorrhiza is a symbiotic association between fungi and plant roots.

21) A unicellular organism is found in stagnant water, showing photosynthetic activity in light and predatory behaviour in dark. It has two flagella and a flexible body.
Identify the organism and its key classification trait.

(1) *Euglena* – Pellicle and mixotrophic nature

(2) *Paramoecium* – Cilia and gullet

(3) *Amoeba* – Pseudopodia

(4) *Plasmodium* – Infectious stage in life cycle

22) In a genetic lab, a fungal species with exogenous conidia, septate hyphae, and sac-like asci is isolated. Which species could it be?

(1) *Rhizopus*

(2) *Penicillium*

(3) *Mucor*

(4) *Ustilago*

23) Match the protists with their correct group:

A	Diatoms	1	Euglenoids
B	*Gonyaulax*	2	Dinoflagellates
C	*Trypanosoma*	3	Flagellated protozoan
D	*Euglena*	4	Chrysophyte

(1) A–4, B–2, C–3, D–1

(2) A–1, B–3, C–2, D–4

(3) A–2, B–1, C–4, D–3

(4) A–3, B–2, C–1, D–4

24) **Assertion (A):** Mycorrhiza is beneficial to both plant and fungus.
Reason (R): The plant provides water and minerals to the fungus.

(1) Both A and R are true and R is correct explanation.

(2) A is true, R is false.

(3) Both statement are false.

(4) A is false, R is true.

25) Which characteristic makes diatoms one of the most ecologically significant protists?

(1) Ability to fix nitrogen

(2) Chief producer in ocean ecosystems

(3) Gritty cell wall for filtration

(4) Occurrence of flagella in vegetative stage

26) Why are viruses placed outside the five kingdom classification?

(1) They are too small

(2) They reproduce only sexually

(3) They lack a cellular structure

(4) They show locomotion

27) Which group helps in nutrient cycling and decomposition?
A. Eubacteria
B. Deuteromycetes
C. Chrysophytes
D. Cyanobacteria

(1) A, B, C

(2) A, C, D

(3) A, B

(4) A, B, C, D

28) Which of the following organisms is incorrectly matched with its feature, based on the Five Kingdom classification?

(1) *Nostoc* – Cyanobacterium with heterocyst

(2) *Chlamydomonas* – Unicellular green algae

(3) *Rhizobium* – Nitrogen fixer, Monera

(4) *Mycoplasma* – Cell wall-less, oxygen-dependent organism

29) Which of the following organisms, placed under Kingdom Protista, were earlier misclassified under both Plant and Animal kingdoms respectively ?

(1) *Euglena* and *Amoeba*

(2) *Chlamydomonas* and *Paramoecium*

(3) *Anabaena* and *Plasmodium*

(4) *Mycoplasma* and *Nostoc*

30) Match the organism with its habitat and special feature:

I	Halophiles	A	Marshy area	1	Salt-loving
II	Methanogens	B	Hot springs	2	Produce methane
III	Thermoacidophiles	C	Extreme salt	3	Heat + acid tolerance

Choose the correct matching:

(1) I–C–1; II–A–2; III–B–3

(2) I–B–3; II–C–1; III–A–2

(3) I–A–2; II–B–3; III–C–1

(4) I–C–2; II–A–1; III–B–3

31) **Assertion (A):** *Mycoplasma* can survive without oxygen and are the smallest living cells.
Reason (R): They have rigid cell walls that protect them in anaerobic environments.

(1) Both A and R are true, and R is the correct explanation.

(2) Both A and R are true, but R is not the correct explanation.

(3) A is true, but R is false.

(4) A is false, but R is true.

32) Which one of the following pairs is correctly matched with its cell covering component?

(1) *Euglena* – Cellulose

(2) Diatom – Silica

(3) *Gonyaulax* – Chitin

(4) *Paramoecium* – Plasmodium

33) Which of the following correctly describes the basis of grouping organisms under Protista?

(1) All aquatic and unicellular eukaryotes.

(2) All microscopic and photosynthetic organisms.

(3) All single-celled eukaryotes, regardless of mode of nutrition.

(4) All eukaryotes that lack a true nucleus.

34) Match the following and choose the correct combination:

Organism	Function
A. *Trypanosoma*	i. Infectious spore like stage
B. *Plasmodium*	ii. Causes sleeping sickness
C. Slime Mould	iii. Plasmodium
D. Diatom	iv. Polishing and filtration industry

(1) A-ii, B-i, C-iii, D-iv

(2) A-ii, B-iii, C-i, D-iv

(3) A-iii, B-ii, C-iv, D-i

(4) A-i, B-ii, C-iv, D-iii

35) A scientist finds a unicellular organism with silica-embedded walls and photosynthetic pigments. Which of the following is the most plausible classification?

(1) Euglenoid

(2) Dinoflagellate

(3) Diatom

(4) Amoeboid protozoan

36) Which group from Protista exhibits dual modes of nutrition depending on environmental conditions?

(1) Chrysophytes

(2) Euglenoids

(3) Dinoflagellates

(4) Sporozoans

37) **Assertion (A):** Rhodophyceae are able to survive at greater depths in oceans.
Reason (R): They contain r-phycoerythrin which absorbs blue light effectively.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

38) Match the following genera with the group to which they belong to:

Column I	Column II
A. *Volvox*	i. Phaeophyceae
B. *Laminaria*	ii. Rhodophyceae
C. *Gelidium*	iii. Chlorophyceae
D. *Ulothrix*	iv. Chlorophyceae

(1) A-iii, B-i, C-ii, D-iv

(2) A-i, B-ii, C-iv, D-iii

(3) A-iii, B-iv, C-i, D-ii

(4) A-i, B-ii, C-iii, D-iv

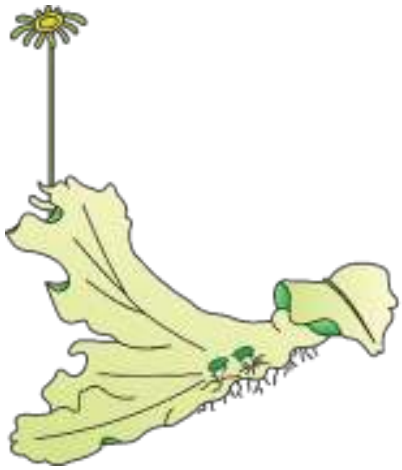
39) Identify the correct statements:
I. *Marchantia* has dorsiventral thallus and gemma cups.
II. Moss gametophyte is independent but sporophyte is photosynthetic and free-living.
III. *Pteridophyte* is ecologically important as a soil binder.
IV. *Selaginella* is homosporous.

(1) I and III only

(2) I, II, III only

(3) I and IV only

(4) I, II and IV only

40) 

Identify the incorrect about above diagram :-

(1) Gemma cup in *Marchantia*

(2) Archegoniophore in female thallus of *Marchantia*

(3) Thalloid gametophyte of *Marchantia*

(4) Rhizoids are multicellular in liverworts

41) Arrange the following events in the life cycle of a fern:
I. Fusion of gametes
II. Formation of prothallus
III. Meiosis in spore mother cell
IV. Spore germination

(1) III → IV → II → I

(2) II → I → IV → III

(3) IV → III → II → I

(4) I → III → II → IV

42) Which of the following features enables gymnosperm leaves (e.g., in *Pinus*) to survive arid climates?

(1) Broad lamina and thin cuticle

(2) Flat leaves with stomata on upper surface

(3) Needle-like leaves with thick cuticle and sunken stomata

(4) Leaves with well-developed chloroplasts only in mesophyll

43) Which of the following statements is/are correct?
I. Pteridophyte gametophyte is dominant and vascularised.
II. Bryophytes have multicellular sex organs and motile sperm.
III. In gymnosperms, female gametophyte develops within ovule.
IV. Angiosperm seeds are naked.

(1) II and III only

(2) I, II, III

(3) I, IV

(4) II, III, IV

44) Which of the following features are common to both gymnosperms and angiosperms?
I. Pollen tube formation
II. Seed formation
III. Fruit formation
IV. Double fertilisation

(1) I, II only

(2) I, II, III

(3) II, III, IV

(4) I, II, IV

45) A botanist discovers a plant with the following features: non-motile male gametes, dominant sporophyte, ovules retained on megasporophyll. The plant is most likely to belong to :-

(1) Moss

(2) Fern

(3) Gymnosperm

(4) Alga

46) **Assertion (A):** Algae contribute significantly to global oxygen production.
Reason (R): Most algae are heterotrophic and parasitic.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

47) Identify the incorrect statement:
(A) *Laminaria* stores food as mannitol and laminarin.
(B) In mosses, gametophyte is the main free-living generation.
(C) Red algae produce motile spores and gametes.
(D) *Chlorella* is a protein-rich unicellular alga.

(1) Only A

(2) Only B

(3) Only C

(4) Only D

48) Arrange the following events in gymnosperm reproduction:
I. Pollen tube growth
II. Pollination
III. Fertilisation

IV. Embryo development

(1) II → I → III → IV
(2) I → II → IV → III
(3) IV → III → II → I
(4) III → IV → I → II

49) In which of the following groups is the sporophyte fully independent and vascularised?

(1) Liverworts
(2) Mosses
(3) Pteridophytes
(4) Red algae

50) Which of the following are features of angiosperms?
I. Double fertilisation
II. Naked seeds
III. Enclosed ovules
IV. Triploid endosperm

(1) I, II, III
(2) I, III, IV
(3) II and III only
(4) I and IV only

51) Match the following plant structures with the group:

Structure	Group
A. Gemma cup	i. Liverwort
B. Prothallus	ii. Pteridophyte
C. Unicellular jacketed sex organs	iii. Algae
D. Microspore	iv. Gymnosperm

(1) A-i, B-ii, C-iii, D-iv
(2) A-ii, B-i, C-iii, D-iv
(3) A-i, B-iii, C-ii, D-iv
(4) A-iv, B-ii, C-i, D-iii

52) Which group shows non-motile female gametes and flagellated male gametes?

(1) Chlorophyceae
(2) Bryophytes
(3) Phaeophyceae
(4) All of the above

53) Which of the following is true about chlorophyceae?
(A) They possess pyrenoids for protein and starch storage.

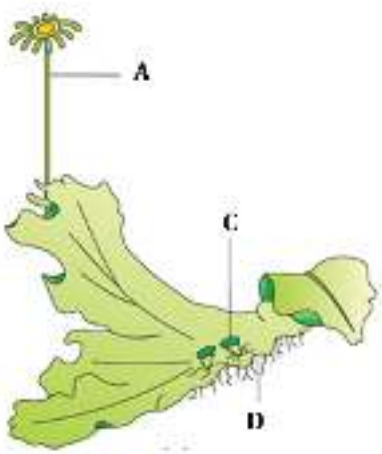
(B) Their chloroplasts are always spiral shaped.
(C) They reproduce only vegetatively.
(D) Their walls are made of lignin.

(1) A only
(2) A and B only
(3) A and C only
(4) B and D only

54) Which evolutionary adaptation is shared by *Selaginella* and gymnosperms but not by ferns?

(1) Xylem with vessels
(2) Heterospory
(3) Flowers
(4) Multicellular gametophyte

55)



(i)



(ii)

Identify the correctly matched labels in the diagram of *Marchantia* (female and male thallus).
Choose the correct combination:

(1) (A) Archegoniophore – Female thallus, (B) Antheridiophore – Male thallus
(2) (A) Antheridiophore – Male thallus, (B) Archegoniophore – Female thallus
(3) (C) Capsule – Male thallus, (B) Seta – Male thallus
(4) (D) Rhizoids – On capsule, (B) Archegonia – On male thallus

56) Which of the following statements are correct regarding Phaeophyceae (brown algae)?
I. Food is stored as laminarin or mannitol.
II. Gametes are biflagellate and of unequal size.
III. Zoospores are pear-shaped and biflagellate
IV. Cell wall is made up of cellulose only.

(1) I and II only
(2) I, II and III only
(3) II, III and IV only
(4) All are correct

57) Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R :**
Assertion A : The first stage of gametophyte in the life cycle of moss is protonema stage.
Reason R : Protonema develops directly from spores produced in capsule.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.

(2) **A** is correct but **R** is not correct.

(3) **A** is not correct but **R** is correct.

(4) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

58) Match the following plant groups with their characteristics:

A	Rhodophyceae	1	Pyrenoids with protein and starch
B	Chlorophyceae	2	Flagella absent in gametes
C	Phaeophyceae	3	Presence of fucoxanthin
D	*Marchantia*	4	Gemma cups for asexual reproduction

(1) A–3, B–1, C–2, D–4

(2) A–2, B–1, C–3, D–4

(3) A–2, B–3, C–1, D–4

(4) A–1, B–3, C–2, D–4

59) Which of the following statements are TRUE for Pteridophytes?
A. They are the first vascular plants on land.
B. Gametophytes are dominant in life cycle.
C. Megaspores and microspores are produced in all pteridophytes.
D. Fertilisation is water-dependent.

(1) A and D

(2) A, B and C

(3) A, C and D

(4) All of the above

60) In which plant group(s) is the gametophyte stage entirely dependent on sporophyte for nutrition and not free-living?

(1) Algae and Pteridophytes

(2) Bryophytes and Angiosperms

(3) Gymnosperms and Angiosperms

(4) Gymnosperms only

61) Which of the following statements is correct for *Selaginella*?

(1) *Selaginella* is homosporous

(2) It is homosporous & heterosporous both
(3) Microspores produce male gametophytes; megaspores produce female gametophytes
(4) Gametes are non-motile

62) Which statement is not correct about liverworts?

(1) Gametophyte dominant
(2) Motile sperm require water for fertilization
(3) Sporophyte dependent
(4) Gemmae are produced by sporophytes

63) Which of the following adaptations is typical of bryophytes?

(1) Presence of cuticle to reduce water loss.
(2) True roots, stems, and leaves.
(3) Alternation of generations with dominant sporophyte.
(4) Sperm motility requiring water for fertilization.

64) The main difference between artificial and natural classification of plants is that:

(1) Artificial systems consider internal features, natural systems do not.
(2) Artificial systems are based on one or few superficial morphological character, natural systems consider gross morphology.
(3) Artificial systems use embryology and phytochemistry.
(4) Natural systems use only gross morphology.

65) Select the incorrectly matched pair -

(1) *Selaginella* - Produce two kind of spores
(2) *Sphagnum* - Provide peat that have long been used as fuel
(3) *Gelidium* - Sexual reproduction is isogamous type
(4) *Fucus* - Gametes are pyriform

66) Which is not a character of cartilaginous fishes?

(1) 2- Chambered heart
(2) Placoid scales
(3) Air bladder
(4) Gills without operculum

67) Diploblastic and triploblastic are terms that describe

(1) The number of invaginations during embryonic development
(2) The number of heads during embryonic development
(3) The number of germinal layers during embryonic development
(4) The number of cell types during development

68) Metagenesis means :-

(1) Polyp produce medusae asexually and medusae produce polyp sexually
(2) Polyp produce medusae sexually and medusae produce polyp sexually
(3) Polyp produce medusae asexually and medusae produce polyp asexually.
(4) Both (1) & (2)

69) Match list-I with list-II

	List-I		**List-II**
(A)	Contractile Vacuole	(I)	Asteries
(B)	Water vascular system	(II)	Spongilla
(C)	Canal system	(III)	Fasciola.
(D)	Flame cells	(IV)	Amoeba

Choose the correct answer from the option given below :-
(1) A–IV, B–I, C–II, D–III
(2) A–I, B–III, C–II, D–IV
(3) A–III, B–II, C–I, D–IV
(4) A–II, B–I, C–III, D–IV

70) The name 'amphibian' indicates that they can live in

(1) Aquatic habitat
(2) Terrestrial habitat
(3) Both (1) and (2)
(4) Aerial habitat

71) Find out incorrect match from column-I, II and III :-

Column-I		Column-II	Column-III
(1)		*Ornithorhynchus*	oviparous, cloaca absent
(2)		*Macropus*	Marsupium present, yolk sac, placenta present
(3)		*Pteropus*	Flying fox, viviparous
(4)		*Balenoptera*	Blue whale, Muscular diaphragm

(1) 1
(2) 2

(3) 3

(4) 4

72)

How many of the following animals shows organ/organ system level of body organisation along with bilateral symmetry ?

Pleurobrachia, *Fasciola*, *Ancylostoma*, *Hirudinaria*, *Anopheles*, adult *Cucumaria*, *Balanoglossus*

(1) Five

(2) Four

(3) Two

(4) Three

73) Match the column I, II and III and choose the correct combination from the options given below :-

	Column I (Figure)		Column II (Type of Coelom)		Column III (Example)
a.		1.	Pseudocoelomate	Q.	Fasciola
b.		2.	Acoelomate	R.	Asterias
c.		3.	Coelomate	S.	Roundworm

(1) a–3–R, b–1–S, c–2–Q

(2) a–2–R, b–3–S, c–1–Q

(3) a–3–S, b–1–Q, c–2–R

(4) a–1–R, b–2–S, c–3–Q

74) Read the following statements with respect to Aves:-
(i) Respiration is through skin.
(ii) The fore limbs always possess scales.
(iii) Long bones are pneumatic.
(iv) Heart is completely three chambered.
(v) Digestive tracts have additional chambers called crop and gizzard.
Which of the above statements are correct?

(1) (iii), (v)

(2) (iii), (iv), (v)

(3) (i), (ii)

(4) (i), (ii), (iii)

75)

Which of the following statements are **correct** about notochord ?
(i) It is derived from mesoderm
(ii) It is formed on the ventral side
(iii) In nonchordate notochord is ventral, double and solid.
(iv) *Balanoglossus* have notochord in their embryonic stage
(v) In vertebrates notochord gives rise to vertebral column

(1) i, iii, v

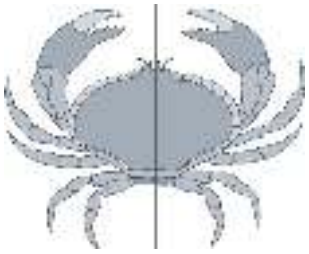
(2) i, iv, v

(3) i, v

(4) i, iii, iv, v

76)

Which of the following is correct option about given diagram?



	Phylum	**symmetry**
(1)	Coelentrata	Bilateral
(2)	Arthropoda	Bilateral
(3)	Arthropoda	Radial
(4)	Mollusca	Radial

(1) 1

(2) 2

(3) 3

(4) 4

77) Select the correct statements with respect to cyclostomes:
(A) They have 6-15 pairs of gill slits for respiration
(B) They have circular sucking mouth with jaws
(C) Their body is devoid of scales and paired fins.
(D) They have cartilaginous vertebral column
(E) They migrate to deep sea for spawning
Choose the **most appropriate** answer from the options given below:

(1) A, B and D only
(2) A, C and D only
(3) A, C and E only
(4) All except E

78) In which of the following animals, hollow and pneumatic long bones are present?

(1) *Balaenoptera, Balanoglossus, Bangarus*
(2) *Catla, Corvus, Columba*
(3) *Pavo, Psittacula, Corvus*
(4) *Corvus, Columba, Chameleon*

79)

Match **Column-I** with **Column-II** and select the **correct** option using the codes given below :-

Column-I		**Column-II**	
a	Tailed Amphibian	i	*Vipera*
b	Flightless bird	ii	*Salamander*
c	Limbless reptile	iii	*Ichthyophis*
d	Limbless amphibian	iv	*Struthio*

Codes:

	a	**b**	**c**	**d**
(1)	iii	iv	ii	i
(2)	ii	iv	i	iii
(3)	iii	i	iv	ii
(4)	i	ii	iii	iv

(1) 1
(2) 2
(3) 3
(4) 4

80) The members of subphylum .....A..... possess .....B..... during embryonic period and replaced by .....C..... in the adult. Choose correct words and fill the blank space A, B, C.

(1) Urochordata; Notochord; Vertebral Column
(2) Vertebrata; Notochord; Vertebral Column
(3) Cephalochordata; Vertebral Column; Jaws
(4) Euchordata; Agnatha; Notochord

81)

Match the column :-

Column-I		Column-II	
(A)	Bufo	(i)	Tree frog
(B)	Rana	(ii)	Limbless-amphibia
(C)	Hyla	(iii)	Frog
(D)	Ichthyophis	(iv)	Toad

(1) A – (iv), B – (iii), C – (i), D – (ii)

(2) A – (iv), B – (i), C – (iii), D – (ii)

(3) A – (ii), B – (iv), C – (i), D – (iii)

(4) A – (i), B – (ii), C – (iv), D – (iii)

82) Which among these is the correct combination of viviparous animals ?

(1) *Macropus,* Platypus*, Camelus*

(2) *Platypus, Camelus, Macaca*

(3) *Camelus, Macaca, Macropus*

(4) *Macaca, Camelus, Platypus*

83) The radial symmetry is observed in
I. Platyhelminthes
II. Coelenterates
III. Aschelminthes
IV. Annelids
V. Echinoderms

(1) I, II, III and V

(2) II, III and I

(3) I, III and IV

(4) II and V

84) Match the following classes with the correct vertebrate group :

	Column-I		Column-II
(a)	Amphibia	(i)	Cartilaginous skeletons
(b)	Aves	(ii)	Hair, mammary glands
(c)	Chondrichthyes	(iii)	Bony skeleton
(d)	Mammalia	(iv)	Pneumatic bones

(1) a-iii, b-iv, c-i, d-ii

(2) a-i, b-iii, c-ii, d-iv

(3) a-ii, b-iv, c-i, d-iii

(4) a-iv, b-ii, c-i, d-iii

85) How many points are correct about the diagram given below ?



(a) Member of phylum- Mollusca
(b) Member of largest phylum
(c) Coelom absent
(d) Bilaterally symmetrical
(e) Triploblastic
(f) Paired jointed appendages

(1) Four
(2) Five
(3) Six
(4) Three

86)

Select the correct option with respect to characteristics of each group :-

	Cyclostomes	**Chondrichthyes**	**Osteichthyes**
(i)	Marine	Marine, Fresh water	Marine
(ii)	Sucking mouth	Ventral mouth	Terminal mouth
(iii)	6-15 Pairs of Gills	5-7 Pairs of Gills	4 Pairs of Gills
(iv)	Placoid scales	Cycloid Scales	Ctenoid/Ganoid scales

(1) (i), (ii) are correct
(2) (i), (iii), (iv) are correct
(3) (ii), (iii) are correct
(4) (ii), (iii), (iv) are correct

87) How many are bony fishes in given table ?
Lamprey, Trygon, Pristis, Labeo, Magur, Sea horse, Dogfish, Ichthyophis

(1) 3
(2) 5
(3) 2
(4) 7

88) A student brought home a strange animal which he found outside under a rock. It had moist skin, a complete digestive tract, a ventral nerve cord, and have metameric segments. Identify the phylum of the animal.

(1) Porifera
(2) Annelida
(3) Mollusca
(4) Echinodermata

89) Bilaterally symmetrical, triploblastic and acoelomate animals are :-

(1) Ctenophore
(2) Platyhelminthes
(3) Aschelminthes
(4) Mollusca

90) Metameric segmentation is present in :-

(1) Annelida, Arthropoda, Mollusca
(2) Arthropoda, Mollusca, Echinoderms
(3) Aschelminthes, Arthropoda, Echinoderms
(4) Annelida, Arthropoda, Chordata

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	3	3	2	4	2	1	3	3	4	2	4	3	4	3	4	3	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	2	2	4	3	3	1	3	2	1	1	3	3	3	1	2	1	2	3
Q.	41	42	43	44	45															
A.	1	2	1	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	1	1	3	3	4	1	4	1	4	3	2	2	2	2	1	3	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	3	3	3	2	2	1	3	4	4	2	1	1	1	1	1	1	4	2
Q.	86	87	88	89	90															
A.	2	1	1	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	3	4	4	3	1	4	1	1	2	1	3	3	1	2	1	1	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	1	2	2	3	3	4	2	1	3	2	3	1	3	2	1	1	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	1	1	3	3	3	1	3	2	1	4	1	2	1	2	4	2	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	4	2	3	3	3	1	1	3	1	1	1	1	3	2	2	3	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	4	1	4	3	1	2	2	4										

SOLUTIONS

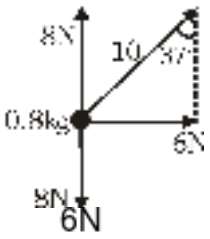
PHYSICS

1) **Explanation :** Find the time in which stone raised.
Concept : This question is based on tension force
Solution :
The maximum tension = 30 × 10 N = 300 N
T – mg = ma
300 – 10 × 10 = 10 a
⇒ a = 20 m/s^2 [Maximum value]
when the mass is raised, u = 0, a = 20 m/s^2
s = 10 m, t = ?
$10 = \frac{1}{2} \times 20 \times t^2$ ⇒ t = 1 second
Final Answer : (2)

2) **Explanation :** Find the contact time of ball with bat
Concept : This question is based on average force and impulse
Solution :
Velocity by which the ball hits the bat
$v_1 = \sqrt{2gh_1} = \sqrt{2 \times 10 \times 5}$ or v_1 = 10 m/s
Velocity of rebound
$v_2 = \sqrt{2gh_2} = \sqrt{2 \times 10 \times 5}$ or v_2 = – 20m/s
$F = m\frac{dv}{dt} = \frac{m|(\vec{v}_2 - \vec{v}_1)|}{dt} = \frac{0.4|-20-10|}{dt} = 100N$
 dt = 0.12 second
Final Answer : (1)

3) **Explanation :** Find the tension in the rope
Concept : This question is based on tension force
Solution :
At 11th second lift is moving upward with acceleration
$a = \frac{v_f - v_i}{\Delta t} \Rightarrow a = \frac{0-3.6}{2} = -1.8 m/s^2$
Tension in rope,
T = m(g – a) = 1500 (9.8 – 1.8) = 12000 N.
Final Answer : (3)

4) **Explain the Question :** We need to find the acceleration of the cabin and the angular position, θ, from the data given.
Concept : This question is based on frames of reference and pseudo force.
Solution :



$a = \overline{0.8} = 7.5\ m/s^2$; $\theta = 37°$

Final Answer : 3

5)

Explain the Question : A particle of **mass m = 2.0 kg** is initially acted upon by **five forces** (F_1 to F_5), resulting in an acceleration of **$4m/s^2$ in the east direction**. When force F_1 is removed, the acceleration becomes **$7m/s^2$ in the north direction**. find the **acceleration of the block if only F_1 is acting** on it.

Concept : This question is based on Newton's Second Law (Vector Form)
Formula : $F_{net} = ma$

Solution :

$\vec{F}_{net} = m\vec{a}$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 2 \times 4\hat{i}$ ....(1)

$\vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 2 \times 7\hat{j}$ ...(2)

eq (1) – eq (2)

$\vec{F}_1 = 2\left(4\hat{i} - 7\hat{j}\right)$

$m\vec{a} = 2(4\hat{i} - 7\hat{j})$

$2(\vec{a}) = 2(4\hat{i} - 7\hat{j})$

$\vec{a} = 4\hat{i} - 7\hat{j} \Rightarrow |\vec{a}| = \sqrt{4^2 + 7^2} \Rightarrow a = \sqrt{65}\ m/s^2$

Final answer : Option (2)

6) **Explanation :** Find the no. of bullets fired per second
Concept : This question is based on average force
Solution :

Let n be the maximum number of bullets per second the person can fire. $F = \frac{nmv}{t}$

$\Rightarrow 144 = n \times \frac{40}{100} \times 1200$

$\Rightarrow n = 3$

Final Answer : (4)

7) **Explanation :** Find interaction force
Concept : This question is based on Newton' second law
Solution :
Let F be the upward force, then
5ma = F – 5 mg
or F = 5m(a + g)
Let the force of interaction between top (first) link and second link be F_{12}. The forces on the

top link are
Upward force F, weight (mg) and
Elastic force of interaction F_{12} by which the remaining four links pull the top link downward.
Thus, $ma = F - mg - F_{12}$
$\Rightarrow F_{12} = F - mg - ma$
$F_{12} = 5m(a + g) - m(a + g) = 4m(a + g)$
$= 4 \times 0.1(2.5 + 9.8) = 4.92$ N
Final Answer : (2)

8) **Explanation :** Find the acceleration of the block A will be (g = 10 m/s)
Concept : This question is based on two blocks problem (smooth surface)
Solution :
Limiting value of static frictional force on B = $\mu_s m_B g$
$f_L = 0.5 \times 7 \times 10 = 35$ N
AS F = 100 N
Hence, $F < f_L$
and the two bodies will move in the same direction, (i.e., of applied force), but with different accelerations.
$\Rightarrow$ Kinetic friction $f_k = m_k\, m_b\, g$
for body A, $f_k = m_a a_A$

$$a_A = \frac{f_k}{m_A} = \frac{\mu_k m_B g}{m_A} = \frac{0.4 \times 9 \times 10}{35} = \frac{28}{35}$$

$a_A = 0.8$ m/s
Final Answer : (1)

9) **Explanation :** Find the minimum mass of A must be
Concept : This question is based on pulley - block system with spring
Solution :
The spring will exert maximum force when the ball is at its lowest position. If the ball has descended through a distance x to reach the position.

$$mgx = \frac{1}{2}Kx^2 \qquad (1)$$

For the block B to leave constant spring force
$Kx = Mg$ (2)
Comparing Eqs. (1) and (2), $m = \frac{M}{2}$
Final Answer : (3)

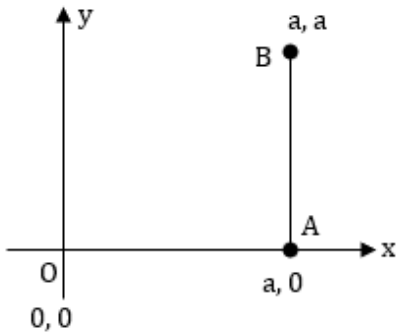
10) **Explanation :** Select correct a-t graph
Concept : This question is based on motion on rough horizontal plane
Solution :
Block starts sliding over the surface when, F max force of friction
$Kt = \mu mg$
or $t = \frac{\mu mg}{K}$
For $t > \mu mg/K$:
net acceleration of block is

$$a = \frac{F - \mu mg}{m} = \frac{Kt}{m} - \mu g$$

Which is a straight line with positive slope and negative intercept. So for $t \le \mu mg/K$, acceleration of the particle is zero and for $t > \mu mg/K$, acceleration is $a = Kt/m - \mu g$
Final Answer : (3)

11) **Explanation :** Find work done by force
Concept : This question is based on work - energy theorem
Solution :
$s = 6t^3 - t^2 - 1 \quad \Rightarrow v = \frac{ds}{dt} = 18t^2 - 2t$
At $t = 0$, $v = 0$
At $t = 3$ s, $v = 18 \times 9 - 2 \times 3 = 156$ m/s
Using work-energy theorem
$$W = \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2} \times 3(156^2 - 0) = 36504J$$
Final Answer : (4)

12) **Explanation :** Find total work done by the force
Concept : This question is based on work done by variable force
Solution :
Given $\vec{F} = -\hat{k}(y\hat{i} + x\hat{j})$
and $\vec{r} = \hat{i}x + \hat{j}y$ So that $d\vec{r} = \hat{i}dx + \hat{j}dy$
Now work done is
$$W = \int \vec{F}d\vec{r} = -\int \hat{k}(y\hat{i} + x\hat{j})(\hat{i}dx + jdy)$$
Let the work done from O to A is W_1 and from A and B is W_2.
For path 0 to A :
$$w_1 = \int_{(0,0)}^{(a,0)} -K(ydx + xdy)$$
along line on the x-axis
$y = 0$, which means $dy = 0$
$w_1 = 0$



For the path A to B, $x = a$ and $y = a$, therefore
$W_2 = -ka \cdot a = -ka^2$
Total work done $W = W_1 + W_2 = -ka^2$
Final Answer : (2)

13) **Explanation :** Find the work done by the force in 2 second is
Concept : This question is based on work done by constant force

Solution :

$x = \frac{t^2}{4} \Rightarrow v = \frac{dx}{dt} = \frac{t}{2} \Rightarrow a = \frac{dv}{dt} = \frac{1}{2} m/s^2$

$\therefore F = ma = 6 \times \frac{1}{2} = 3N$

Displacement of the body in 2 seconds is

$s = 0 + \frac{1}{2} \times \frac{1}{2} \times (2)^2 = 1m$

Work done = F · s = 3 × 1 = 3J

Final Answer : (4)

14) **Explanation :** find nature of force

Concept : This question is based on conservation and non conservative

Solution :

Here, the force is tangential to the curve, hence the work done depends on path. Therefore, force $\vec{F}$-must be non-conservative

Final Answer : (3)

15) **Explanation :** Find the energy stored in Q is

Concept : This question is based on spring potential energy

Solution :

Let the spring constant of spring Q is k and that of P is 2k. The extensions produced by applying equal forces on them are x_P and x_Q, respectively

Since F = Fx (numerically)

$\Rightarrow x = \frac{F}{k}$ and $U = \frac{1}{2}kx^2$

$\Rightarrow U = \frac{1}{2}k\left(\frac{F}{k}\right)^2 = \frac{F^2}{2k} \Rightarrow U \propto \frac{1}{k}$

Thus, $\frac{U_Q}{U_P} = \frac{k_p}{k_Q} = \frac{2k}{k} = 2$ $\Rightarrow U_Q = 2U_P = 2E$

Final Answer : (4)

16) **Explanation :** Find the magnitude of the force on the particle and the acceleration of the particle are

Concept : This question is based on equilibrium

Solution :

$Fx = -\frac{\partial U}{\partial x} = 7$ and $F_y = -\frac{\partial U}{\partial x} = -24$

$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(7)^2 + (-24)^2} = 25N$

$\therefore a = \frac{F}{m} = \frac{25}{5} = 5m/s^2$

Final Answer : (3)

17) **Explanation :** Find the x,y, & z components of force

Concept : This question is based on $F = -\Delta U$ and equilibrium

Solution :

$U = 40 + 6x^2 - 7xy + 8y^2 + 32z$

$\Rightarrow F_x = \frac{-\partial U}{\partial y} = -12x + 7y = 24N$ [Putting x = –2; y = 0]

and $F_y = \frac{-\partial U}{\partial y} = -7x + 16y = -14N$ [Putting x = –2; y = 0]

and $F_z = \frac{-\partial U}{\partial y} = -32N$

Final Answer : (4)

18) **Explanation :** Find the power of the motor

Concept : This question is based on power

Solution :

Total energy drawn from the motor

$= \frac{1}{2}mv^2 + mgh$

$= \frac{1}{2} \times 10 \times 10^2 + 10 \times 10 \times 20$

= 500 + 2000 = 2500 J

$\Rightarrow P = \frac{energy}{time} = \frac{2500}{1} = 2500W = 2.5kW$

(The mass of 1 litre water is 1 kg because density of water is 10^{-3} kg/m^3)

Final Answer : (3)

19) **Explanation :** Find the maximum speed

Concept : This question is based on COME

Solution :

The speed of the particle is maximum when its kinetic energy is maximum and from energy conservation law, the potential energy must be maximum at this point for minimum value of potential energy

$\frac{dV}{dx} = 0$

$\frac{d}{dx}\left[\frac{x^4}{4} - \frac{x^2}{2}\right] = 0$

$\Rightarrow x^3 - x + 0 \Rightarrow x(x^2 - 1) = 0$

$\Rightarrow x = 0$ and $x = \pm 1$

Now $V(\text{at}_{x=0}) = 0$ and $V_{(x=\pm 1)} = \left(\frac{1}{4} - \frac{1}{2}\right) = -\frac{1}{4}J$

Thus, the minimum possible value of potential energy is $-\frac{1}{4}J$

Since total energy = Kinetic energy + Potential Energy

$\Rightarrow$ KE = TE – PE = $2 - \left(-\frac{1}{4}\right) = \frac{9}{4}J$

$\Rightarrow \frac{1}{2}mv^2 = \frac{9}{4} \Rightarrow \sqrt{\frac{9}{2m}} = \sqrt{\frac{9}{2 \times 1}} = \frac{3}{\sqrt{2}}m/s$

Final Answer : (1)

20) **Explanation :** Find the maximum momentum of block

Concept : This question is based on conservation of mechanical energy

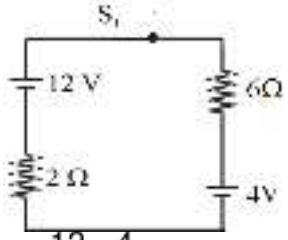
Solution :
Loss in KE of the block = Gain in elastic PE of the spring
$\Rightarrow \frac{1}{2}Mv^2 = \frac{1}{2}KL^2 \Rightarrow M^2v^2 = MkL^2$
$\Rightarrow Mv = \sqrt{Mk}\,L$
Final Answer : (3)

21) **Explanation :** Find the increase in kinetic energy
Concept : This question is based on work energy theorem
Solution :
$(\text{speed})_i = u = \frac{p_i}{m} = \frac{8}{4} = 2\text{m/s}$
Impulse = increase in momentum
Ft = mv – mu
$\Rightarrow 0.2 \times 10 = 4[v - 2]$
$\Rightarrow v = \frac{2}{4} + 2 = 2.5\text{m/s}$
$\therefore \Delta K = \frac{1}{2}m(v^2 - u^2) = \frac{1}{2} \times 4[(2.5)^2 - (2)^2]$
= 2[6.25 – 4] = 4.5 J
Final Answer : (3)

22) **Explanation :** Find the power delivered to the particle
Concept : This question is based on power
Solution :
$a_r = \frac{v^2}{r} = k^2rt^2 \quad \therefore v^2 = k^2r^2t^2$
$\Rightarrow K = \frac{1}{2}mv^2 = \frac{1}{2}mk^2r^2t^2$
Using work-energy theorem,
$W = \Delta K = \frac{1}{2}mk^2r^2t^2 - 0$
$\therefore P = \frac{\Delta K}{\Delta t} = mk^2r^2t$
Final Answer : (3)

23) **Explaining :**
Given : Two charged capacitors are connected.
Ask : Final charge on capacitors
Concept : This question is based on Key :
(1) Conservation of charge
(2) Final potential across capacitor will be same.
Solution :
charge on A = $\frac{C_1}{C_1 + C_2}Q_{\text{Total}} = 6q$
charge on B = $\frac{C_2}{C_1 + C_2}Q_{\text{Total}} = 3q$
Final Answer : (2)

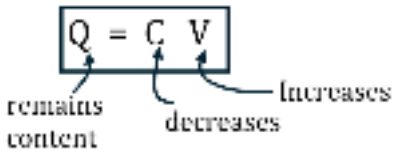
24) **Explaining :**
Given : RC circuit with two switches S_1 and S_2.
Ask : Steady state current in 6Ω resistance.
Concept : This question is based on Key : In steady state capacitor is fully charged and acts as open circuit.
Solution :



$$I = \frac{12-4}{2+6}$$
$$= \frac{8}{8} = 1A$$
Final Answer : (2)

25)

Explanation : Find A parallel plate capacitor is charged using a battery. After the capacitor is fully charged, the battery is disconnected and the plates are moved apart
Concept : This question is based on As the battery is disconnected, charge on the capacitor (Q) remains constant.
Solution :
$\Rightarrow \sigma = \left(\frac{Q}{A}\right)$ remains constant
$\Rightarrow$ $\boxed{\text{Electric field} = \left(\frac{\sigma}{A}\right) \text{ remains constant.}}$
Note : $C = \frac{A\varepsilon_0}{d}$
d (Increases) $\Rightarrow$ C (decreases)



Final Answer : (4)

26)

Explain Question : The battery remains connected to the capacitor, when separation between plates in increased, we need to comment, what happens to, q, v, u and E.

Concept : This question is based on parallel plate capacitor (q, c, v, u, E)

Solution :
$(\downarrow c) = \frac{A\in_0}{(d \uparrow)}$
$(\downarrow q) = (\downarrow c)$ (v) ; v is constant as battery is still connected.

$(\downarrow u) = \frac{1}{2}(\downarrow c)\ (v^2)$

$(\downarrow E) = \frac{(v)}{(d \uparrow)}$

Final Answer : option (3) Stored energy decreases.

27) **Explanation:**
Given : Two Charged capacitor is connected with opposite polarity.
Ask : Final Energy of the configuration.
Concept : This question is based on Key :

(1) $V_{common} = \frac{C_2V_2 - C_1V_1}{C_1 + C_2} = V_c$

(2) $V_{comb} = \frac{1}{2}(C_1 + C_2)V_c^2$

Solution :

$V_{common} = \frac{C_2V_2 - C_1V_1}{C_1 + C_2}$

$V_{common} = \frac{(2C)2V - CV}{2C + C} = V$

$U_{combination} = \frac{1}{2}(C_1 + C_2){V_C}^2$

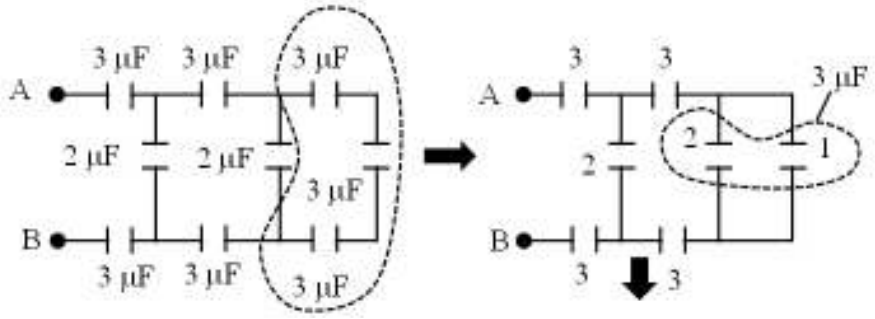
$= (2C + C)V^2 = \frac{3}{2}CV^2$

Final Answer : (3)

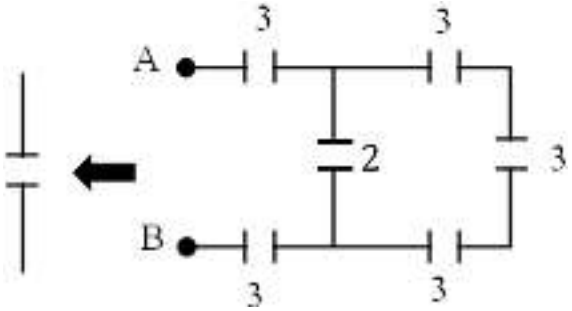
28) **Explanation :** Find equivalent capacitance
Concept : This question is based on Combine capacitors using series and parallel rules, and simplify the circuit systematically to find net capacitance between two terminals.
Solution :
Formula

- Series: $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$
- Parallel: $C_{eq} = \sum C_i$

Calculation:
The given circuit can be redraws as follows



On further solving the network in similar manner equivalent capacitance obtained between A and B will be 1μF.



Final Answer : (1)

29) **Question Explanation :**
We need to find increase in energy of capacitor when its potential is increased, with the help of data given.

Concept :
This question is based on Energy stored in capacitor.

Solution :

$$U = \frac{1}{2}CV^2$$

$$\Delta U = \frac{1}{2}C(V_2^2 - V_1^2)$$

$$\Delta U = \frac{1}{2} \times (6 \times 10^{-6}) \times [(20)^2 - (10)^2]$$

$\Delta U = 3 \times 10^{-6}\ [300]$

$\Delta U = 9 \times 10^{-4}\ J$

Final Answer : (3) : 9×10^{-4} J

30) **Explain Question:**
Asking about the power delivered to patient by a charged capacitor in 2ms.

Concept:
Energy stored in capacitor.

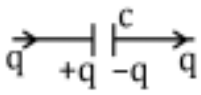
Solution:

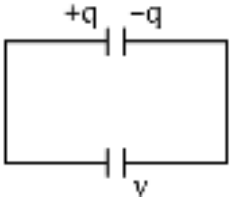
$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{\frac{1}{2}CV^2}{t} = \frac{(40 \times 10^{-6})(3000)^2}{2 \times (2 \times 10^{-3})}$$

$= 90$ kW

Final Answer: Option (2)

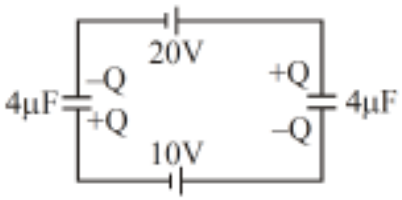
31) **Explaining :**
Given : A capacitive circuit
ASK : Charge on particular capacitor
Concept : This question is based on Key :

(1) Charge distribution 



(2) KVL : $V - \frac{q}{C} = 0$

Solution :



using KVL in close loop

$-\frac{Q}{4} - 10 - \frac{Q}{4} + 20 = 0$

$\frac{Q}{2} = 10 \Rightarrow Q = 20\ \mu C$

Final Answer : (1)

32) **Explaining :** Two sources of emf 6 V and internal resistance 3 Ω and 2 Ω are connected to an external resistance R as shown. If potential difference across battery A is zero, then value of R is :

Concept : This question is based on $i = \frac{E_{net}}{R_{net}}$.

Solution :

$i = \frac{E_{net}}{R_{net}} = \frac{6+6}{3+2+R} = \frac{12}{5+R}$

$(V)_A = E_A - irA = 0$

$E_A = ir_A$

$6 = \left(\frac{12}{5+R}\right)(3) \Rightarrow 12 = 10 + 2R$

$R = 1\Omega$

Final Answer : (1)

33) **Explanation:** Find the separation of a parallel plate capacitor made of a square and a circular plate. The capacitance depends on the smaller, circular plate's area.

Concept: Capacitance of a parallel plate capacitor, $C = \epsilon_0 A/d$.

Solution:

Using common Area

$C = \frac{\varepsilon_0 \left(\pi \frac{a^2}{4}\right)}{d}$

$d = \frac{\pi \varepsilon_0 a^2}{4C}$

Final Answer : (3)

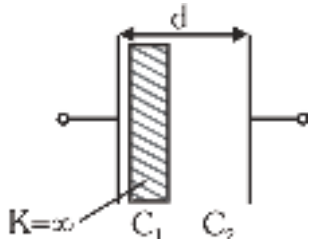
34) **Explanation:** Find the ratio of new to old capacitance (C'/C) when a copper slab of

thickness d/3 is inserted into a parallel plate capacitor.

Concept: A conductor acts as a dielectric with K = ∞ . The new capacitance is given by C' = $\frac{A\in_0}{d-t+t/K}$.

Solution :

$C = \frac{A\varepsilon_0}{d}$



$C' = \frac{A\varepsilon_0}{d-t+\frac{t}{K}}$

$K_{metal} = \infty$

$C' = \frac{A\varepsilon_0}{d-\frac{d}{3}} = \frac{3}{2}C$

Final Answer : (3)

35)

Explanation: Find the equivalent capacitance between points A and B for a system of five identical parallel plates connected as shown in the figure.

Concept: Combination of parallel plate capacitors in series and parallel.

Formula Used:

• Capacitance of a parallel plate capacitor:

$C = \frac{\varepsilon_o A}{d}$

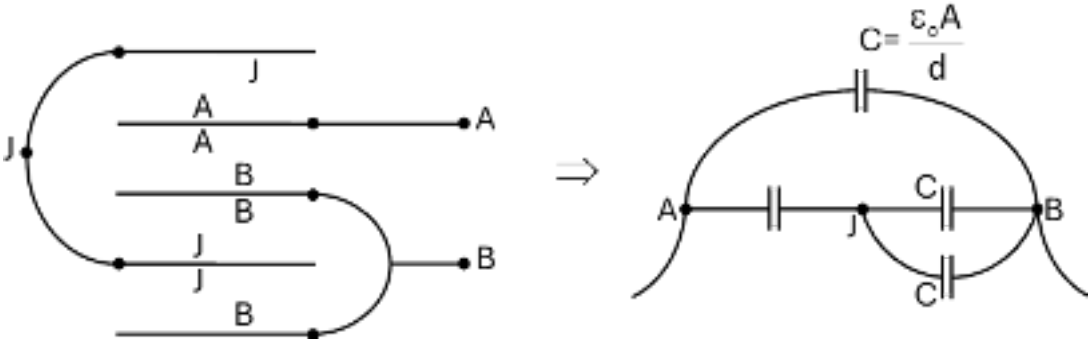
• For series capacitors:

$\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$

• For parallel capacitors:

$C_{eq} = \sum C_i$

Visual Diagram:



Calculation:

$C_{eq} = C + \left(\frac{(C)(2C)}{C+2C}\right)$

$$C_{eq} = C + \frac{2}{3}C$$

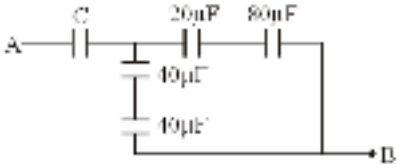
$$C_{eq} = \frac{5}{3}C \quad \Rightarrow C = \frac{\varepsilon_0 A}{d}$$

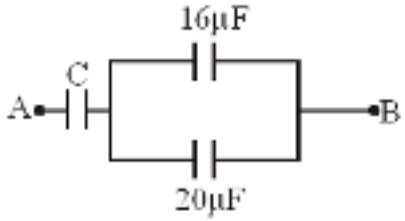
$$\boxed{C_{eq} = \frac{5}{3}\frac{\varepsilon_0 A}{d}}$$

Final Answer:

The correct option is (3).

36) **Explanation:** Find the value of an unknown capacitor C in a network where the effective capacitance between A and B is 12μF.
Concept: Solving capacitor networks using series and parallel combination rules.
Solution:
Equivalent diagram





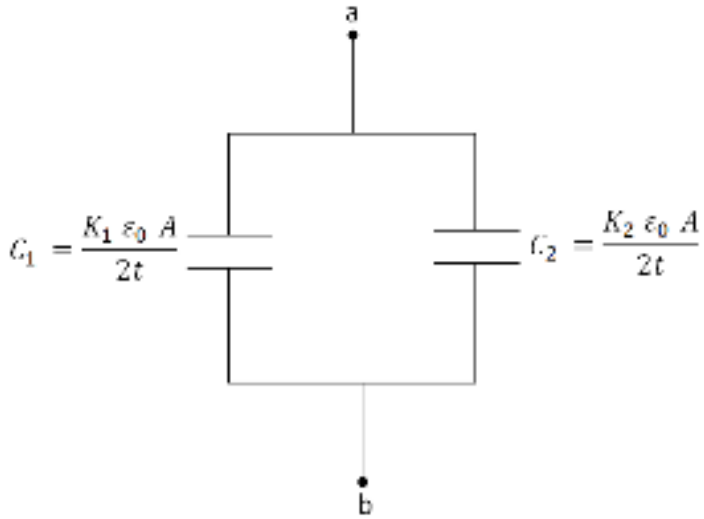
$\equiv$

$$\equiv \frac{C \times 36}{C + 36} = 12$$

C = 18 μF
Final Answer : (1)

37) **Explain Question:** Asking about the effective capacitance for the given arrangement of dielectric.

Concept: Effective capacitance with dielectric.
Solution: The effective capacitance will be of parallel combination



$C_{eq} = C_1 + C_2$

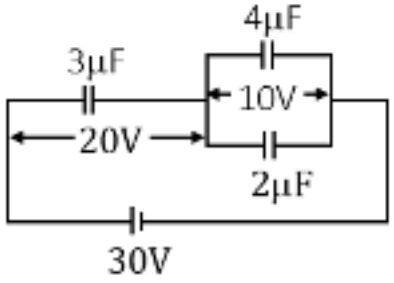
$$C_{eq} = \frac{K_1 \in_0 A/2}{t} + \frac{K_2 \in_0 A/2}{t}$$

$$= \frac{\in_0 A}{t} \cdot \frac{K_1 + K_2}{2}$$

Final answer: option (2)

38) **Explanation:** Calculate the potential difference $V_A - V_B$ for a given circuit segment at a particular instant.
Concept: Apply Kirchhoff's Voltage Law (KVL) or potential drops to find the potential difference across the circuit elements.
Solution : $V_A - 6 - 2 \times 3 + \frac{9 \times 10^{-6}}{1 \times 10^{-6}} - 3 \times 3 = V_B$
$V_A - 6 - 6 + 9 - 9 = V_B$
$V_A - V_B = 12$ Volt
Final Answer : (1)

39) **Explanation:** For the circuit shown, match the charge supplied by the battery and the charges on specific capacitors.
Concept: Capacitors in series have the same charge, and capacitors in parallel have the same voltage. The total charge is $Q = C_{eff} V$.
Solution:



(i) $Q_{battery} = C_{eff} V$
$= 2 \times 30 = 60$ μC
(ii) $Q_{4\mu F} = 4 \times 10$
$= 40$ μC
(iii) $Q_{2\mu F} = 2 \times 10$
$= 20$ μC
(iv) $Q_{3\mu F} = 3 \times 20$
$= 60$ μC
Final Answer : (2)

40) **Explanation:** Find the amount of charge on a capacitor in a DC circuit.
Concept: At steady state in a DC circuit, a capacitor acts as an open circuit, and no current flows through its branch. The charge is Q = CV where V is the potential difference across the capacitor.
Solution :

$$I = \frac{V}{R_{net}} = \frac{6}{2.5 + 0.5} = 2A$$

$V_{2\Omega} = IR = 1 \times 2 = 2$ V

$V_{5\mu F} = 2$ V
$Q = CV = 5 \times 2 = 10$ μC
Final Answer : (3)

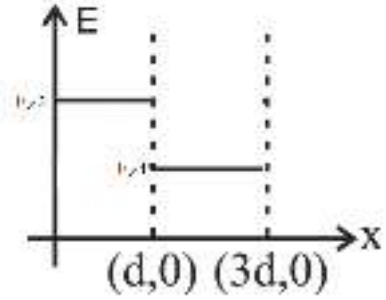
41) **Explain Question:**
Asking about the correct graph between electric field and distance, for given dielectric arrangement.

Concept:
Electric field in dielectric.

Solution:
Electric field between plates of capacitor is given by,

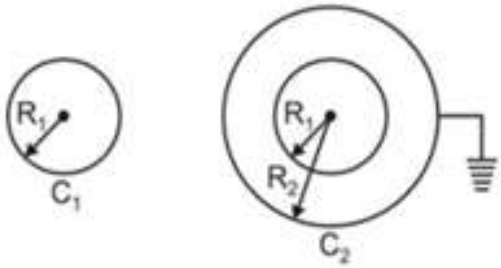
$$E = \frac{\sigma}{k\in_0}$$

(K → Dielectric constant)
Or, we can say $E_{medium} = \frac{E_0}{K}$
then,



Final answer: option (1)

42) **Explaining :** To find the radius of spherical shell.
Concept : This question is based on spherical capacitor.
Solution :



$C_1 = 4\pi\varepsilon_0 R_1$
$C_2 = 4\pi\varepsilon_0\left(\frac{R_1R_2}{R_2-R_1}\right)$
And
Given that $C_2 = nC_1$

$$\frac{R_1R_2}{R_2-R_1} = nR_1$$

$$\frac{R_2/R_1}{\frac{R_2}{R_1}-1}$$

$$\frac{R_2}{R_1} = \frac{n}{n-1}$$

Final Answer : (2)

43) **Explaining :** To find equivalent capacitance.

Concept : This question is based on combination of spherical capacitor.

Solution : Consider two spherical capacitors in series ;

$C_1 = \frac{4\pi\varepsilon_0 ab}{b-a}$ and $C_2 = \frac{4\pi\varepsilon_0 bc}{c-b}$



$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{\left(\frac{4\pi\varepsilon_0 ab}{b-a}\right)\left(\frac{4\pi\varepsilon_0 bc}{c-b}\right)}{4\pi\varepsilon_0 b\left[\frac{a}{b-a} + \frac{c}{c-b}\right]}$$

$$\Rightarrow C_{eq} = \frac{4\pi\varepsilon_0 abc}{a(c-b) + c(b-a)} = \frac{4\pi\varepsilon_0 abc}{ac - ab + cb - ac}$$

$$= \frac{4\pi\varepsilon_0 ac}{c-a}$$

Final Answer : (1)

44) **Explaining :** To find the potential drop across the plates of capacitor.

Concept : This question is based on RC circuit.

Solution : $I = \frac{2V - V}{3R} = \frac{V}{3R}$

Apply loop law in upper loop.

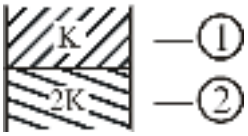
We get voltage across capacitor = $\frac{V}{3}$

Final Answer : (3)

45) **Explaining :** Choose the correct option when dielectrics are introduced bet. the plates of capacitor.

Concept : This question is based on parallel plate capacitor.

Solution :



$V_1 = V_2$

$E_1 d = E_2 d$

$E_1 = E_2$

$C_2 = 2C_1$

$U_1 = \frac{1}{2} C_1 V_1^2 = \frac{1}{2}$ Kcv^2

$U_2 = \frac{1}{2} C_2 V_2^2 = \frac{1}{2}$ (2Kc)v^2

$U_2 > U_1$

Final Answer : (2)

46)

Explaining : The correct order of ionic radii for the following ions is :
Concept : This question is based on ionic bond
Solution :
Across a period, as the nuclear charge increases, the size of the ions generally decreases. Therefore, the ionic radii decrease from left to right: $Ti^{3+} > V^{3+} > Cr^{3+} > Mn^{3+}$. The correct order from smallest to largest is $Mn^{3+} < Cr^{3+} < V^{3+} < Ti^{3+}$.
Final Answer : (3)

47)

Explaining : Lanthanoid contraction is a consequence of :
Concept : This question is based on Lanthanoid contraction
Solution :
The lanthanoid contraction is the steady decrease in atomic and ionic radii of the lanthanoids with increasing atomic number. This is due to the poor shielding effect of the 4f electrons, which results in a greater effective nuclear charge pulling the outer electrons closer to the nucleus.
Final Answer : (2)

48)

Explaining : The general electronic configuration of d-block elements is:
Concept : This question is based on general configuration
Solution :
The general electronic configuration for d-block elements is $(n-1)d^{1-10}ns^{1-2}$. The ns orbital fills before the $(n-1)$d orbital, and there are some exceptions like Cr and Cu where the ns electron is shifted to the $(n-1)$d orbital for stability.
Final Answer : (1)

49)

Explaining : Which of the following ions is colorless
Concept : This question is based on colorless
Solution :
The color of transition metal ions is generally due to d-d transitions. Sc^{3+} has an electronic configuration of $[Ar]^3d^0$. Since there are no d-electrons, no d-d transition is possible, and the ion is colorless.
Final Answer : (1)

50)

Explaining : Which of the following statements about d-block elements is incorrect
Concept : This question is based on general properties of d-block
Solution :

Due to their ability to show multiple oxidation states and form complexes, d-block elements and their compounds (e.g., V_2O_5, Fe, Ni) are widely used as catalysts. Therefore, the statement that they are generally poor catalysts is incorrect.
Final Answer : (3)

51)

Explaining : Which of the following is the most stable ion in an aqueous solution
Concept : This question is based on stability of O.S.
Solution :
The stability of transition metal ions in aqueous solution is often related to their electronic configuration. Fe^{3+} has a stable half-filled d^5 configuration, making it particularly stable.
Final Answer : (3)

52)

Explaining : large positive value of $E^{\circ}_{M^{3+}/M^{2+}}$

Concept : This question is based on large positive value of $E^{\circ}_{M^{3+}/M^{2+}}$
Solution : Mn^{2+} has half filled (d^5) stable configuration due to which it has large M^{3+} / M^{2+} value.
Final Answer : (4)

53) **Explaining :** Variable oxidation states is:
Concept : This question is based on variable oxidation states is:
Solution :
Scandium (Z = 21) does not exhibit variable oxidation states.
Ni, Ti and Co show +2, +3 and +4 oxidation states.
Final Answer : (1)

54) **Explaining :** Color in ions
Concept : This question is based on color in ions
Solution :

Configuration	Aquated ion	Colour
$3d^0$ (No unpaired electron)	Sc^{3+}	Colourless
$3d^{10}$ (No unpaired electron)	Zn^{2+}	Colourless
$3d^0$ (No unpaired electron)	Ti^{4+}	Colourless
$3d^2$ (Two unpaired electrons)	V^{3+}	Green

Final Answer : (4)

55) **Explaining :** Oxidation state shown by Mn
Concept : This question is based on oxidation state shown by Mn
Solution : MnF_7 does not exist.
Final Answer : (1)

56) **Explaining :** O.S. of Mn
Concept : This question is based on O.S. of Mn
Solution : Due to multiple bond ability of oxygen.
Final Answer : (4)

57) **Explaining :** Electronic configuration of Cm
Concept : This question is based on electronic configuration of Cm
Solution : Electronic configuration of Cm is [Rn] $5f^7$ $5d^1$ $7s^2$
Final Answer : (3)

58)

Explaining : Which of the following compounds has a three-dimensional polymeric structure
Concept : This question is based on polymer
Solution :
Silicon dioxide (SiO_2) forms a continuous tetrahedral network, giving it a three-dimensional polymeric structure. The others are either discrete molecules or different structural types.
Final Answer : (2)

59)

Explaining : The correct order of Lewis acidic strength for the boron halides is:
Concept : This question is based on back bonding
Solution :
The Lewis acidity increases as the extent of $p\pi-p\pi$ back-bonding decreases. The back-bonding is most effective in BF_3 (due to small size of F) and least effective in BI_3, making BI_3 the strongest Lewis acid.
Final Answer : (2)

60)

Explaining : Which of the following allotropes of carbon is used as a lubricant?
Concept : This question is based on allotrope of carbon
Solution :
Graphite has a layered structure with weak van der Waals forces between the layers. These layers can slide over one another, which makes it an excellent lubricant.
Final Answer : (2)

61)

Explaining : The correct order of thermal stability of the hydrides of Group 16 elements is:
Concept : This question is based on thermal stability
Solution :
As we move down the group, the E-H bond length increases and the bond dissociation enthalpy decreases. This leads to a decrease in thermal stability. Thus, H_2O is the most stable.
Final Answer : (2)

62)

Explaining : In which of the following compounds does fluorine exhibit an oxidation state other than -1
Concept : This question is based on O.S.
Solution :
The oxidation state of any element in its elemental form is zero. For all other given compounds, fluorine, being the most electronegative element, will have an oxidation state of -1.
Final Answer : (1)

63)

Explaining : The most common oxidation state of the elements of Group 13 is:
Concept : This question is based on electronic configuration
Solution :
The valence shell electronic configuration of Group 13 elements is ns^2np^1. They typically lose all three valence electrons to achieve a +3 oxidation state.
Final Answer : (3)

64)

Explaining : Which of the following hydrides of Group 15 elements is the strongest reducing agent
Concept : This question is based on thermal stability
Solution :
The reducing character of the hydrides of Group 15 elements increases down the group because the M—H bond strength decreases, making the hydrogen atom easier to donate.
Final Answer : (3)

65)

Explaining : The oxidation state of phosphorus in hypophosphorous acid (H_3PO_2) is:
Concept : This question is based on oxyacids
Solution :
Let the oxidation state of P be x. The sum of oxidation states is zero.
$3(+1) + x + 2(-2) = 0$
$3 + x - 4 = 0$
$x = +1$
Final Answer : (1)

66)

Explaining : Which of the following represents the correct order of bond dissociation enthalpy for halogens
Concept : This question is based on bond dissociation energy.
Solution :
The bond dissociation enthalpy generally decreases down the group. However, due to the small size of fluorine, there is significant inter-electronic repulsion between the lone pairs, which weakens the F—F bond. This makes its bond enthalpy anomalously low, lower than that of Cl_2 and Br_2.
Final Answer : (2)

67)

Explaining : Which of the following is not an oxoacid of phosphorus
Concept : This question is based on oxyacid
Solution : Peroxodisulphuric acid ($H_2S_2O_8$) is an oxoacid of sulfur, not phosphorus.
Final Answer : (4)

68) **Explaining :** Which statement about molecularity of a reaction is wrong ?
Concept : This question is based on molecularity.
Solution : Molecularity can never be fractional.
Final Answer : (3)

69) **Explaining :** Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is:
Concept : This question is based on Arrhenius equation.
Solution : $k = Ae^{-E_a/RT}$ i.e., k increase exponentially, with rise in temperature.
Final Answer : (3)

70) **Explaining :** Which of the following statement is correct for a reaction X + 2Y $\longrightarrow$ Products?
Concept : This question is based on rate of appearance and disappearance.
Solution : $-\frac{d[X]}{dt} = -\frac{1}{2}\frac{d[Y]}{dt} = \frac{d[P]}{dt}$.
Final Answer : (3)

71) **Explaining :** Rate law expression.
Concept : This question is based on rate law for complex reactions.
Solution : For slow step, rate = $k[O][O_3]$ ...(i)
Also for equilibrium $K_c = \frac{[O][O_2]}{[O_3]}$ ...(ii)
$[O] = \frac{K_c[O_3]}{[O_2]}$ ...(iii)
By Eqs. (iii) and (i), the intermediate [O] is eliminated as

rate = $\frac{k \cdot K_C[O_3][O_3]}{[O_2]} = \frac{K'[O_3]^2}{[O_2]}$

Final Answer : (2)

72) **Explaining :** Determine order.

Concept : This question is based on pseudo order reaction.

Solution :

The reactant takes in excess obeys zero order reaction;

 r = k [B].

Final Answer : (2)

73) **Explaining :** For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, the $\frac{d[NH_3]}{dt} = 2 \times 10^{-4} Ms^{-1}$.

are Therefore, the rate $-\frac{d[N_2]}{dt}$ is given as:

Concept : This question is based on rate of reaction.

Solution : $\frac{-d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

Final Answer : (1)

74) **Explaining :** The rate constant k_1 and k_2 for two different reactions are $10^{16}\ e^{-2000/T}$ and $10^{15}\ e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is :

Concept : This question is based on Arrhenius equation.

Solution :

$K_1 = 10^{16}\ e^{-2000/T}$; $K_2 = 10^{15}\ e^{-1000/T}$

if $K_1 = K_2$ then $10^{16}\ e^{-2000/T} = 10^{15}\ e^{-1000/T}$

or $\log 10 - \frac{2000}{T} = \frac{1000}{T}$ or $T = \frac{1000}{2.303}K$

Final Answer : (3)

75) **Explaining :** Half-life period of a first order reaction is 1386 seconds. The specific reaction rate of the reaction is :

Concept : This question is based on first order reactions.

Solution : For first order, $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{1386} = 5 \times 10^{-4} s^{-1}$

Final Answer : (4)

76) **Explaining :** Determine rate law.

Concept : This question is based on rate law.

Solution : Use : $r = k [A]^m[B]^n$

Final Answer : (4)

77) **Explaining :** Evaluate statement and explanation.

Concept : This question is based on collision theory.

Solution : Only those molecules react which acquire threshold energy level & have proper orientation

Final Answer : (2)

78) **Explaining :** In a first order reaction A → B B, if k is the rate constant and initial concentration of the reactant is 0.5 M, then half-life is :
Concept : This question is based on first order reaction.
Solution :
For I order reaction, $k = \frac{2.303 \log_{10} 2}{t_{1/2}}$
$\square$ $t_{1/2} = \frac{0.693}{k} = \frac{\ln 2}{k}$
Final Answer : (1)

79) **Explaining :** Activation energy of a chemical reaction can be determined by :
Concept : This question is based on Arrhenius equation.
Solution : By 2.303 $\log \frac{K_2}{K_1} = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$
Final Answer : (1)

80)

Explaining : Find ppm(w/w)
Concept : This question is based on concentration term.
Solution : $150 \times 10^3 \rightarrow 450 \times 10^{-3}$

$$\Rightarrow 10^6 \rightarrow \frac{450 \times 10^{-3}}{150 \times 10^3} \times 10^6 = 3 \text{ ppm}$$

Final Answer : (1)

81)

Explaining : Find P_A = ?
Concept : This question is based on Raoult's law.
Solution : $P_A^0 = 75$ torr, $P_B^0 = 22$ torr
P_A = ? $w_A = 78$ g $w_B = 46$

$$P_A = P_A^0 X_A \Rightarrow 75 \times \left(\frac{1}{1 + \frac{1}{2}}\right) = 50 \text{ torr}$$

Final Answer : (1)

82)

Explaining : Calculate vapour pressure of solution ?
Concept : This question is based on Raoult's law
Solution : V. P. of solution = $X_{\text{heptane}} \times 92 + X_{\text{octane}} \times 31$

$$= \frac{1}{5} \times 92 + \frac{4}{5} \times 31$$
$$= 43.2 \text{ mm of Hg}$$

Final Answer : (1)

83) **Question Explanation :** The question asks the correct scientific definition of a solution.
Concept : Definition of solution.
Solution : Homogeneous mixture of two or more substances is called as solution.
Final Answer : homogeneous mixture of two or more substances.

84) **Explaining :** Correct statements among the following are
Concept : This question is based on concentration terms
Solution :
(a) Commercial bleaching solution contains 3.62 mass percentage of sodium hypochlorite in water: Standard liquid household bleach is a dilute solution of sodium hypochlorite (NaOCl). Commercial formulations are typically standardized at **3.62%** by mass (w/w), meaning 100 grams of solution contains 3.62 grams of the active bleaching agent.
(b) 35% (v/v) solution of ethylene glycol, an antifreeze, is used in cars for cooling the engine: Ethylene glycol is the primary component of automotive antifreeze. A 35% volume/volume (v/v) aqueous solution is commonly used because it effectively lowers the freezing point of water to approximately 255.4 K (–17.6°C), protecting the engine in cold climates.
(c) Sum of mole fraction of all components in a solution is one: By definition, the mole fraction (x) of a component is $X_i = \frac{n_i}{n_{total}}$. Since the total number of moles in a mixture is the sum of the moles of all individual components, the sum of all mole fractions must always equal **unity (1)**.
(d) Mass % and mole fraction are independent of temperature: Concentration terms that depend solely on mass (like mass percentage, mole fraction, and molality) do not change with temperature. In contrast, terms involving volume (like molarity) are temperature-dependent because volume expands or contracts as temperature changes.
Final Answer : (4)

85) **Explaining :** 0.59 gm. of a dibasic acid is present in 100 c.c. of $\frac{N}{10}$ solution. What is the molecular weight of the acid ?
Concept : This question is based on normality
Solution : 100 c.c. of $\frac{N}{10}$ acid Now we have : Normality = $\frac{w}{\text{eqvt. wt.}} \times \frac{1000}{\text{Volume (c.c.)}}$

$\frac{1}{10} = \frac{0.59}{\text{eqvt. wt.}} \times \frac{1000}{100}$ eqvt. wt. = 59 gm. ∴ Molecular weight = basicity × eqvt. wt. = 2 × 59 = 118 **Final Answer :** (2)

86) **Explaining :** In osmosis phenomenon net flow of :
Concept : This question is based on osmosis
Solution : As per definition of osmosis
Final Answer : (2)

87) **Explain question:**
We must find the total vapour pressure of a solution made from 2 moles of A and 3 moles of B, using Raoult's law.
Concept:
Total vapour pressure:

$P = x_A \cdot P_A^\circ + x_B \cdot P_B^\circ$

Solution:

Total moles = 2 + 3 = 5

$x_A = 2/5 = 0.4$

$x_B = 3/5 = 0.6$

$P = 0.4 \times 100 + 0.6 \times 150$

$P = 40 + 90 = 130$ mm Hg

Final Answer: Option (1)

88)

Explaining the Question :- Find total vapor pressure expression for binary ideal solution using Raoult's law.

Concept:- This question is based on Raoult's law.

Solution :-

For ideal solution $P_{total} = P_A + P_B = P_A^0 X_A + P_B^0 X_B$

substitute $X_A = 1 - X_B$.

$P_{total} = P_A^0(1 - X_B) + P_B^0 X_B = P_A^0 + (P_B^0 - P_A^0)X_B$

Final Answer :- 1

89) **Explain the Question :** The question asks to identify which pair of solutions will exert the same osmotic pressure (be isotonic) when kept at the same temperature.

Concept : This question is based on osmotic pressure.

Solution :

$\pi = CRT$

For isotonic solutions :

at same temp $\Rightarrow \pi_1 = \pi_2$

$C_1 = C_2$

Option (2) : 3.42 g can sugar (sucrose)

$$C_1 = \frac{\text{Mol}}{\text{L}} = \frac{3.42}{342} = 0.01 \text{ Mol L}^{-1}$$

0.18 g glucose :

$$C_2 = \frac{\text{Mol}}{\text{L}} = \frac{0.18}{180} = 0.01 \text{ Mol L}^{-1}$$

$C_1 = C_2 \Rightarrow$ Isotonic solutions.

Final Answer : (2)

90)

Question Explanation

The question asks for the freezing point of an aqueous solution of glucose ($C_6H_{12}O_6$) given the mass of glucose, the mass of water, and the molal depression constant of water.

Concept

The freezing point depression (ΔT_f)

Explanation

$$\Delta T_f = K_f m = 1.86 \times \left(\frac{18}{180} \times \frac{1000}{1000}\right) = 0.186$$

$T_f^0 - T_f = 0.186$, $T_f = -0.186°C$
Final Answer (1)

BIOLOGY

91) **Solution:** A defining property of life must be present in all living organisms without exception and absent in non-living objects. Consciousness (ability to sense external surroundings and respond) is a defining feature of all living organisms. Growth and reproduction have exceptions, and isolated metabolic reactions outside the body in a cell-free system are not living organisms.
Final Answer : (3)

92) **Solution:** Response to external stimuli (consciousness) is present in all living organisms without exception, making it a defining characteristic. Non-living things can show external growth (accumulation of material), and several living organisms do not reproduce (e.g., sterile worker bees, mules).
Final Answer : (4)

93) **Solution :** A-III, B-I, C-IV, D-II
A. Growth → III. Increase in mass and increase in number of individuals (Twin characteristics)
B. Reproduction → I. Production of offspring
C. Metabolism → IV. Sum total of all chemical reactions occurring in a living body
D. Cellular organization → II. Composed of one or more living cells
Final Answer : (3)

94) **Solution:** Living organisms uniquely possess metabolic activities, consciousness (sensing surroundings), growth, cellular organisation, and reproduction collectively—features absent in non-living entities. Therefore, all listed options are unique features of life forms.
Final Answer : (4)

95) **Solution:** I**ncrease in body mass** and **increase in number of individuals** are known as the **twin characteristics of growth**. Thus, statements **(i)** and **(iii)** are correct.
Final Answer : (4)

96) **Solution :** Ernst Mayr is known as The Darwin of 20th century.
Final Answer : (3)

97) **Solution:** In plants, cell division and growth occur **continuously** throughout their lifespan via meristematic tissues. In animals, growth occurs **only up to a certain age**, after which cell division occurs mainly to replace lost or damaged cells.
Final Answer : (1)

98) **Solution:** Defining characteristics are features shown by all living organisms without exception. **Metabolism**, **Consciousness**, and **Cellular organization** are all defining features

of life.
Final Answer : (4)

99) **Solution :**
Statement 1 – Correct: An order consists of related families that share a few similar characteristics.
Statement 2 – Incorrect: As we move from species to kingdom, the number of common characteristics **decreases**, not increases.
Statement 3 – Correct: Convolvulaceae and Solanaceae are grouped in Polymoniales mainly based on floral similarities.
Statement 4 – Correct: Carnivora includes families like Felidae (cats) and Canidae (dogs).
Statement 5 – Correct: Classes such as fishes, amphibians, reptiles, birds, and mammals are included in Chordata due to the presence of a notochord and dorsal hollow nerve cord.
Statement 6 – Incorrect: Kingdom is the **highest**, not the lowest, taxonomic category.
Final Answer : (1)

100) **Solution :** (a)–ii, (b)–iii, (c)–iv, (d)–i
• Class → Mammalia
• Order → Carnivora
• Family → Felidae
• Kingdom → Plantae (general example of kingdom used here in match)
Final Answer : (1)

101) **Solution :** Role in shaping present atmosphere.
Final Answer : (2)

102) **Solution :** A. Enzymes are secreted outside the body
B. Digestion occurs extracellularly
C. Digested nutrients are absorbed
Final Answer : (1)

103) **Solution :** Dinoflagellates have a cellulose-based cell wall.
Final Answer : (3)

104) **Solution :** All of the above.
Final Answer : (3)

105) **Solution :** Bacteria are grouped under four categories based on their shape: *Coccus* (spherical), *Bacillus* (rod-shaped), Vibrium (comma-shaped), *Spirillum* (spiral).
Final Answer : (1)

106) **Solution :** The viruses are non-cellular organisms. The protein coat called capsid.
Final Answer : (2)

107) **Solution :** Archaebacteria differ from other bacteria in having a different cell wall

structure. *Mycoplasma* can survive without oxygen.
Final Answer : (1)

108) **Solution :** Fungi had chitin in their walls while green plants had a cellulosic cell wall.
Final Answer : (1)

109) **Solution :** A–3, B–1, C–2, D–4
Phycomycetes — Aseptate coenocytic hyphae.
Ascomycetes — Sac-like asci.
Basidiomycetes — Basidiospores on basidia.
Deuteromycetes — Imperfect fungi (no sexual stage).
Final Answer : (1)

110) **Solution :** Penicillium – an Ascomycete.
Final Answer : (3)

111) **Explain Question :** The question asks to identify a unicellular organism found in stagnant water that is photosynthetic in light, predatory in dark, has two flagella, and a flexible body, along with its key classification trait.
Concept : Kingdom Protista – Euglenoids: Biological Classification
Solution : Euglenoids are unicellular protists commonly found in stagnant freshwater. They possess a protein-rich pellicle instead of a cell wall, which makes their body flexible. They have two flagella (one long and one short) and show mixotrophic nutrition: photosynthetic in sunlight and heterotrophic (predatory) in absence of light. The organism described matches these characteristics.
Final answer : (1)

112) **Solution :** *Penicillium* – Ascomycete, exogenous conidia,septate mycelium.
Final Answer : (2)

113) **Solution :** A–4, B–2, C–3, D–1
Diatoms — Chrysophyte
Gonyaulax — Dinoflagellates
Trypanosoma — Flagellated protozoan
Euglena — Euglenoids
Final Answer : (1)

114) **Explain Question:** The question asks to evaluate the correctness of the Assertion and Reason related to mycorrhiza and select the correct option.
Concept: Biological Classification-Mycorrhiza
Solution: Mycorrhiza is a symbiotic association between fungi and roots of higher plants. Both partners are benefited, so the assertion is true. The fungus helps the plant in absorption of water and minerals, especially phosphorus, while the plant provides food to the fungus. Hence, the reason statement is false.
Final Answer: (2)

115) **Solution :** Diatoms are the chief producer in ocean ecosystem, that's why they are ecologically significant.
Final Answer : (2)

116) **Solution :** Viruses did not find a place in classification since they are not considered truly 'living'.
Final Answer : (3)

117) **Solution :** Heterotrophic bacteria – decomposers Deuteromycetes - decomposers.
Final Answer : (3)

118) **Solution :** They are the smallest living cells known and can survive without oxygen.
Final Answer : (4)

119) **Solution :** Kingdom Protista has brought together *Chlamydomonas*, *Chlorella* (earlier placed in Algae within Plants... with *Paramoecium* and *Amoeba*... placed in the animal kingdom).
Final Answer : (2)

120) **Solution :** Extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens). Methanogens responsible for methane production.
Final Answer : (1)

121) **Solution :** The *Mycoplasma* are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen.
Final Answer : (3)

122) **Solution :** In diatoms the cell walls form two thin overlapping shells. The walls are embedded with silica and thus the walls are indestructible.
Final Answer : (2)

123) **Solution :** All single-celled eukaryotes are placed under Protista. Being eukaryotes, the protistan cell body contains a well-defined nucleus and other membrane-bound organelles.
Final Answer : (3)

124) **Solution :** *Trypanosoma* cause diseases such as sleeping sickness.
Plasmodium (malarial parasite) causes malaria and have infectious spore like stage.
Slime moulds form an aggregation called plasmodium.
Diatoms used in polishing, filtration.
Final Answer : (1)

125) **Solution :** In diatoms the cell walls are embedded with silica. Most of them are photosynthetic.

Final Answer : (3)

126) **Solution :** Though they are photosynthetic in the presence of sunlight, when deprived of sunlight they behave like heterotrophs.
Final Answer : (2)

127) **Explaining :**
Concept : Pigment absorption in algae.
Solution : r-phycoerythrin enables red algae to capture low-intensity blue light penetrating deep water.
Final Answer : (1)

128) **Concept :** Algal classification by pigment and storage material.
Solution : *Volvox* and *Ulothrix* belong to Chlorophyceae (green algae), *Laminaria* to Phaeophyceae (brown algae), and *Gelidium* to Rhodophyceae (red algae).
Final Answer : (1)

129) **Concept :** Differentiating bryophyte and pteridophyte structures and spores.
Solution : II is incorrect – moss sporophyte is dependent. IV is incorrect – *Selaginella* is heterosporous.
Final Answer : (1)

130) **Explaining :**
Concept : Bryophyte diagram and structure.
Solution : Rhizoids in liverworts (like *Marchantia*) are unicellular; mosses have multicellular rhizoids.
Final Answer : (4)

131) **Explaining :**
Concept : Alternation of generations in Pteridophytes.
Solution : Sporophyte undergoes meiosis → spores → germinate → prothallus (gametophyte) → gametes fuse.
Final Answer : (1)

132) **Explaining :**
Concept : Leaf adaptations for water conservation.
Solution : Needle-shape, sunken stomata, and thick cuticle reduce transpiration.
Final Answer : (3)

133) **Explaining :**
Concept : Gametophyte roles and seed coverage.
Solution : I and IV are incorrect; sporophyte is dominant in pteridophytes, and angiosperms have enclosed seeds.
Final Answer : (1)

134) **Explaining :**
Concept : Reproductive similarities.
Solution : Fruits and double fertilisation are angiosperm features only.
Final Answer : (1)

135) **Concept :** Characteristics of gymnosperms.
Solution : Retained ovule and non-motile gametes indicate gymnosperm.
Final Answer : (3)

136) **Explaining :**
Concept : Photosynthetic activity in algae.
Solution : Algae are autotrophic and photosynthetic, producing large amounts of oxygen. R is false.
Final Answer : (3)

137) **Explaining :**
Concept : Life cycle and structure in algae and mosses.
Solution : Red algae have non-motile spores and gametes. Statement C is incorrect.
Final Answer : (3)

138) **Explaining :**
Concept : Gymnosperm fertilisation sequence.
Solution : Pollination → pollen tube → fertilisation → embryo.
Final Answer : (1)

139) **Explaining :**
Concept : Alternation of generations and plant body dominance.
Solution : In pteridophytes, the sporophyte is dominant, vascularised and independent.
Final Answer : (3)

140) **Explaining :**
Concept : Unique angiosperm reproductive traits.
Solution : Naked seeds (II) are gymnosperm trait. I, III, IV are correct.
Final Answer : (2)

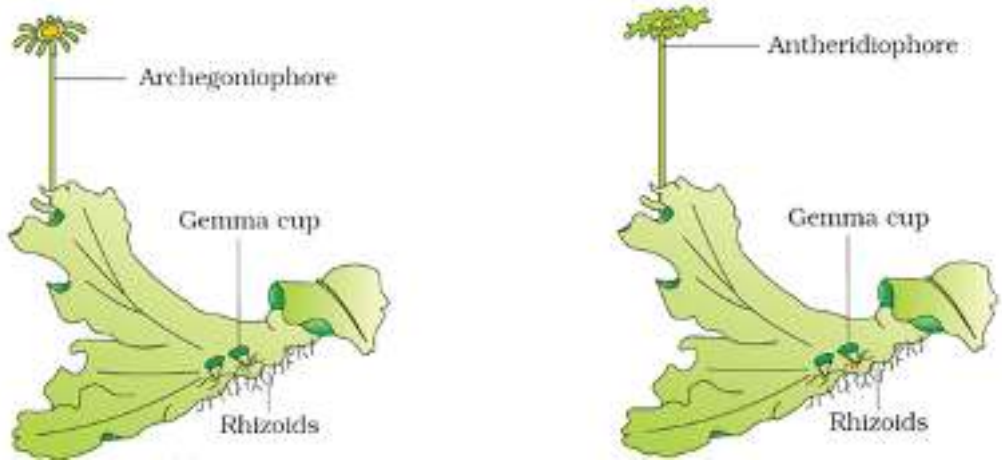
141) **Explaining :**
Concept : Group-specific reproductive structures.
Solution : Gemma in liverworts, prothallus in ferns, archegonia in bryophytes, microspore in gymnosperms.
Final Answer : (1)

142) **Solution :** All listed groups have oogamous reproduction.
Final Answer : (4)

143) **Explaining :**
Concept : Algal ultrastructure.
Solution : A is true. B is false (various shapes); C is false (also asexual and sexual); D is false.
Final Answer : (1)

144) **Explaining :**
Concept : Seed evolution steps.
Solution : Heterospory is a precursor to seed habit, seen in *Selaginella* and gymnosperms.
Final Answer : (2)

145) **Concept :** From NCERT Fig. 3.2 and description of liverworts – *Marchantia* has archegoniophore (female) and antheridiophore (male).



Solution : "Figure 3.2 Bryophytes: A liverwort – Marchantia (a) Female thallus (b) Male thallus".
Final Answer : (1)

146) **Solution :** Statement IV is incorrect — wall has cellulose + outer algin.
The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin.
Final Answer : (2)

147) **Solution :** The first stage of gametophyte in the life cycle of moss is protonema stage because protonema develops directly from spores produced in capsule.
Final Answer : (4)

148) **Solution :**
Rhodophyceae — Flagella absent in gametes.
Chlorophyceae — Pyrenoids with protein and starch.
Phaeophyceae — Presence of fucoxanthin
Marchantia — Gemma cups for asexual reproduction.
Final Answer : (2)

149) **Solution :** B is false (sporophyte is dominant), C is false (only some are heterosporous).
Final Answer : (1)

150) **Solution :** Unlike bryophytes and pteridophytes, in gymnosperms the male and the female gametophytes do not have an independent free-living existence.
Final Answer : (3)

151) **Solution :** *Selaginella* is heterosporous; microspores → male gametophytes, megaspores → female gametophytes. B is not always true; gametes are motile in pteridophytes.
Final Answer : (3)

152) **Solution :** Gemmae are produced by gametophytes.
Final Answer : (4)

153) **Solution :** Bryophytes have motile sperm and require water for fertilization. They lack true roots, stems, leaves, sporophyte is dependent, not dominant.
Final Answer : (4)

154) **Solution :** Artificial systems are based on one or few superficial morphological character, natural systems consider gross morphology.
Final Answer : (2)

155) **Solution :** *Gelidium* - Sexual reproduction is isogamous type
Final Answer : (3)

156) **Solution :** Air bladder is not a character of cartilaginous fishes.
Final Answer : (3)

157) **Question Explanation:** Germ layer terminology
Concept: Number of germinal layers
Solution: Diploblastic and triploblastic describe the number of germinal layers (two or three) present during embryonic development in animals.
Final Answer: The number of germinal layers during embryonic development

158) **Explain Question :** Metagenesis means
Concept : This question is based on metagenesis
Explanation : In Obelia and other cnidarians, polyps reproduce asexually to form medusae, while medusae reproduce sexually to form polyps.
Final answer : 1: Polyp produce medusae asexually and medusae produce polyp sexually

159) **Solution :**

	List-I		**List-II**
(A)	Contractile Vacuole	(IV)	Amoeba
(B)	Water vascular system	(I)	Asteries
(C)	Canal system	(II)	Spongilla

(D)	Flame cells	(III)	Fasciola.

Final Answer : (1)

160) **Solution :** Amphibian indicates that they can live in both Aquatic and Terrestrial habitat
Final Answer : (3)

161) **Solution :**

Column-I		Column-II	Column-III
(1)		*Ornithorhynchus*	oviparous, cloaca present

Final Answer : (1)

162) **Question Explanation:**
This question is asking to count how many animals from the given list possess both organ/organ system level of body organization and bilateral symmetry.
Concept :
Animal Kingdom: Topics – levels of organization and body symmetry in different animal groups.

Explanation:
Analyse each animal:

A. Pleurobrachia – Ctenophore → radial symmetry → NOT counted
B. Fasciola (liver fluke) – Platyhelminth → organ level + bilateral
C. Ancylostoma (hookworm) – Aschelminth → organ system level + bilateral
D. Hirudinaria (leech) – Annelid → organ system level + bilateral
E. Anopheles (mosquito) – Arthropod → organ system level + bilateral
F. Adult Cucumaria (echinoderm) – radial symmetry → NOT counted
G. Balanoglossus – Hemichordate → organ system level + bilateral

Total animals satisfying both conditions = 5

Final Answer: Option 1

163) **Solution :**

	Column I (Figure)		Column II (Type of Coelom)		Column III (Example)
a.		3.	Coelomate	R.	Asterias
b.		1.	Pseudocoelo-mate	S.	Roundworm
c.		2.	Acoelomate	Q.	Fasciola

Final Answer : (1)

164) **Solution :**
(i) Respiration is through lungs.
(ii) The fore limbs modified into wings.
(iv) Heart is completely four chambered
Final Answer : (1)

165) **Solution :**
(ii) It is formed on the dorsal side
(iii) In nonchordate notochord is absent.
(iv) *Balanoglossus* have stomochord not notochord.
Final Answer : (3)

166)

Explain Question : The question asks to identify the correct phylum and symmetry of the given crab diagram.

Concept : This question is based on arthropoda

Solution : The image shows a crab, which belongs to the Phylum Arthropoda and exhibits bilateral symmetry (dividable into two mirror halves along a central plane).

Final Answer: Option (2)

167) **Solution :**
(B) They have circular sucking mouth without jaws
(E) They migrate for spawning to fresh water.
Final Answer : (2)

168) **Solution :** Hollow and pneumatic long bones are present in *Pavo, Psittacula, Corvus*

Final Answer : (3)

169) **Solution :**

Column-I		**Column-II**	
a	Tailed Amphibian	ii	*Salamander*
b	Flightless bird	iv	*Struthio*
c	Limbless reptile	i	*Vipera*
d	Limbless amphibian	iii	*Ichthyophis*

Final Answer : (2)

170) **Solution :** The members of subphylum vertebrata possess notochord during embryonic period and replaced by vertebral Column in the adult.
Final Answer : (2)

171) **Question Explanation:** Match the common names of amphibians with their scientific names.
Concept: Amphibia
Solution: The correct match is A - (iv) Bufo (Toad), B - (iii) Rana (Frog), C - (i) Hyla (Tree frog), D - (ii) Ichthyophis (Limbless amphibian).
Final Answer: 1

172)

Question Explanation: Identify animals that give birth to young.
Concept: Viviparity classification.
Solution: Viviparous animals give birth to live young.
Macropus (Kangaroo) and Camelus (Camel) are viviparous mammals.
Macaca (Monkey) is also a viviparous mammal.
Platypus is an oviparous mammal (lays eggs), so any combination including it is incorrect.
The correct option is 3 which lists all viviparous animals.
Final Answer: (3)

173) **Question Explanation:** Identifying phyla with radial symmetry.
Concept: Radial Symmetry in Animals.
Solution: Radial symmetry is a body plan in which any plane passing through the central axis divides the organism into two identical halves. This type of symmetry is found in Coelenterates and adult Echinoderms. Other options like Platyhelminthes, Aschelminthes, and Annelids exhibit bilateral symmetry.
Final Answer: II and V

174) **Solution :** a-iii, b-iv, c-i, d-ii

	Column-I		**Column-II**

(a)	Amphibia	(iii)	Bony skeleton
(b)	Aves	(iv)	Pneumatic bones
(c)	Chondrichthyes	(i)	Cartilaginous skeletons
(d)	Mammalia	(ii)	Hair, mammary glands

Final Answer : (1)

175) **Solution :**
(a) Member of phylum-Mollusca
(d) Bilaterally symmetrical
(e) Triploblastic
Final Answer : (4)

176)

Explain Question : This question asks about the correct characteristics of Cyclostomes, Chondrichthyes and Osteichthyes.
Concept : This Question is based on Animal Kingdom → Vertebrata → Pisces & Cyclostomata.
Explanation :
(i) Cyclostomes are marine, Chondrichthyes are mostly marine, Osteichthyes occur in both marine and freshwater, not only marine.
(ii) Cyclostomes → sucking mouth; Chondrichthyes → ventral mouth; Osteichthyes → terminal mouth.
(iii) Cyclostomes → 6–15 pairs of gill slits; Chondrichthyes → 5–7 pairs; Osteichthyes → 4 pairs covered by an operculum.
(iv) Cyclostomes → scaleless, not placoid; Chondrichthyes → placoid scales; Osteichthyes → cycloid/ctenoid scales (ganoid in some).
Final answer : - 3

177) **Solution :** Labeo, Magur, and Sea horse are bony fishes.
Final Answer : (1)

178) **Solution :** All characters are related to phylum Annelida.
Final Answer : (2)

179) **Solution :** Platyhelminthes are bilaterally symmetrical, triploblastic and acoelomate animal.
Final Answer : (2)

180)

Question Explanation: Identify the animal groups that show true (metameric) segmentation.

Concept: The question is based on body segmentation

Solution: The correct answer is option 4.

Metameric segmentation means the body is divided into repeated segments externally and internally.

A. **Annelida** (e.g., earthworm): show clear metameric segmentation.
B. **Arthropoda** (e.g., insects): segmentation is present, though often modified.
C. **Chordata** (e.g., vertebrates): show internal metamerism (e.g., vertebrae, muscles).

Mollusca and Echinodermata do not show true metameric segmentation.

Final answer: Option 4